Lecture 23 (11/9)

Regularization

Review of material since midterm 2

1) Using interactions to data where the effect of one variable depends on the others
2) Using residual plots to diagnose a model
3) Building nonlinear models with features, polynomial regression
4) Overfitting and bias-variance tradeoff as model complexity increases

$$
\begin{aligned}
E_D[\hat{\sigma}^2_{\text{test}}] &= E_X\left[\text{MSE}_{\hat{y}(X,D)}\right] + \sigma^2_\epsilon \\
&= \underbrace{E_X\left[\text{var}_D(\hat{y}(X,D))\right]}_{\text{variance}} + \underbrace{E_X\left[(E_D[\hat{y}(X,D)] - f(X))^2\right]}_{\text{bias}^2} + \sigma^2_\epsilon
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Today:

1) Review and practice for unit 5 material
2) Begin discussing regularization, a technique to prevent overfitting

Practice questions: Conceptual

Question

Consider a model $\hat{y} = f(x)$ which predicts the cost of an individual's car insurance based on some attributes $x = (x_1, x_2, x_3, \dots)$. For example, x_1 might be number of previous crashes. Discuss the implications of bias and variance, both from the driver's and the insurance company's perspectives.

Practice questions: Technical

Question

A cancer treatment can use any combination of three drugs A, B and C, where $X_A = 1$ if drug A is used and $X_A = 0$ if not (and the same for X_B and X_C). The response Y is the reduction in tumor size (in mm). The model with all interactions is

$$Y = \beta_A X_A + \beta_B X_B + \beta_C X_C + J_{AB} X_A X_B + J_{AC} X_A X_C + J_{BC} X_B X_C + J_{ABC} X_A X_B X_C + \epsilon$$

with

$$\beta_A = 4, \quad \beta_B = 3, \quad \beta_C = 2,$$
$$J_{AB} = -2, \quad J_{AC} = 1, \quad J_{BC} = -4, \quad J_{ABC} = 2.$$

If drug A is unavailable, what is the best treatment?

Practice questions: Technical

Question

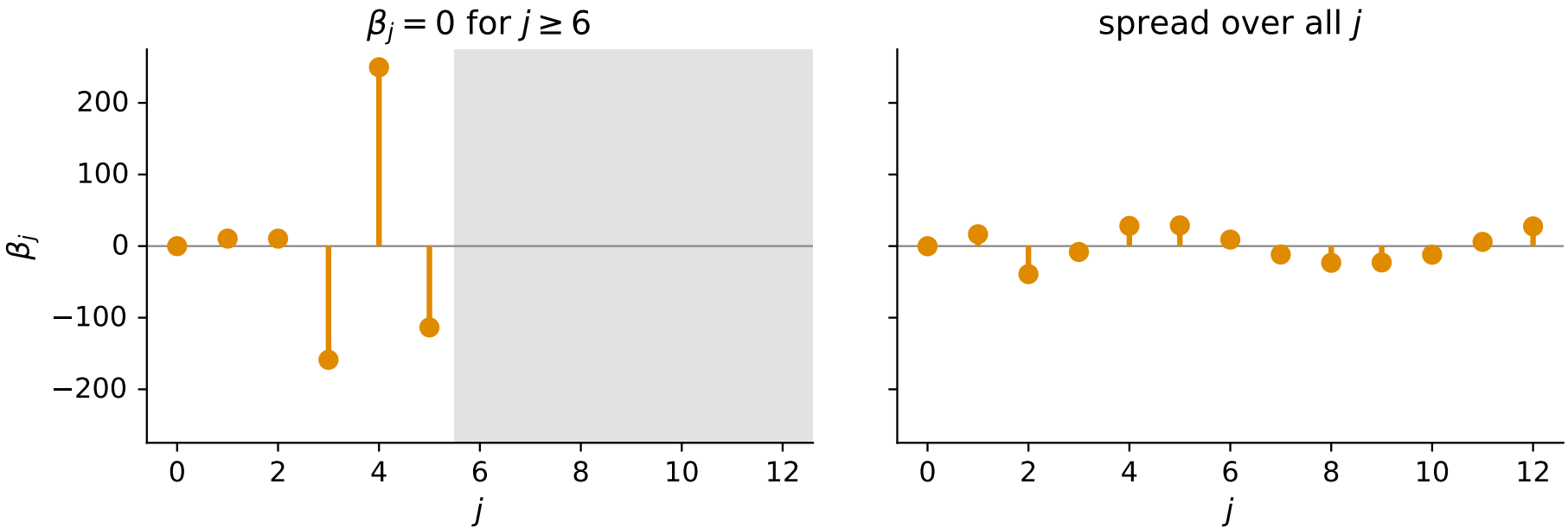
A degree-K polynomial is fit to $N_{\text{train}} = 100$ training points. The training error is $\hat{\sigma}^2_{\text{train}} = 0.5$ and $\hat{\sigma}^2_Y = 2$ for the training Y values. What is R^2 on the training points?

Question

1) Describe how $\hat{\sigma}^2_{\text{train}}$ and $\hat{\sigma}^2_{\text{test}}$ change as K is increased to N_{train}
2) If the model has high variance and low bias in the bias-variance decomposition of $E_D[\hat{\sigma}^2_{\text{test}}]$ does $\hat{y}(x, D^{\text{train}})$ have a large standard error?
3) Can $E_D[\hat{\sigma}^2_{\text{test}}]$ be smaller than σ^2_ϵ?

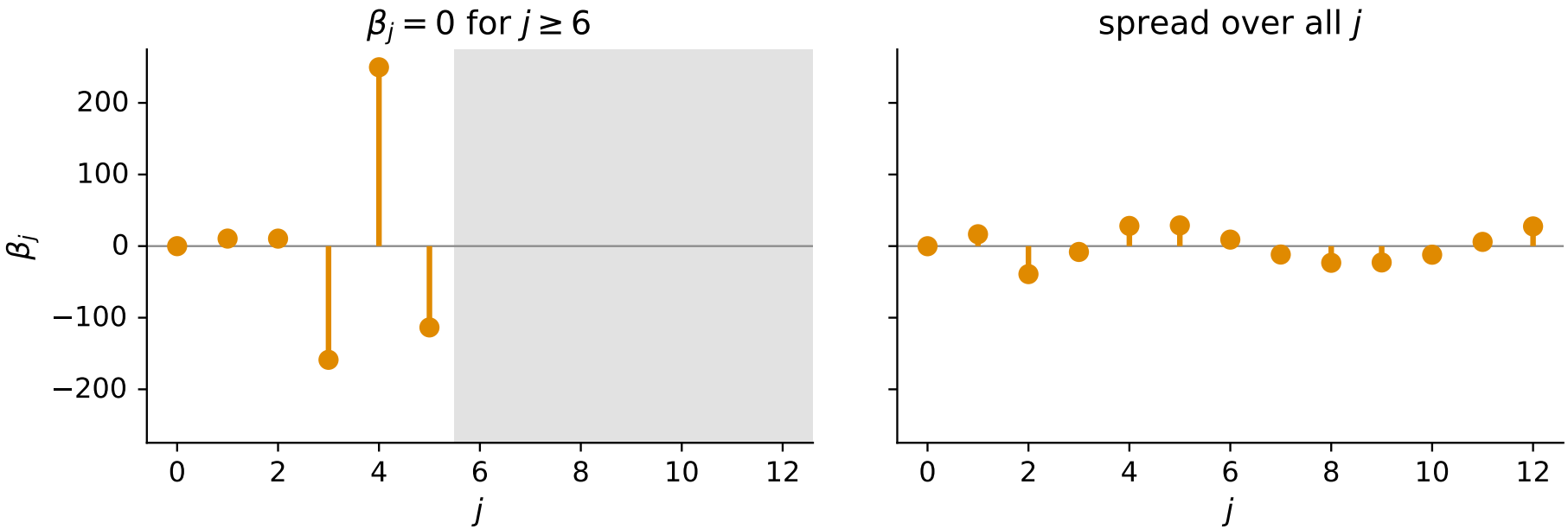
A hard cutoff, or coefficients that shrink?

$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \cdots$$



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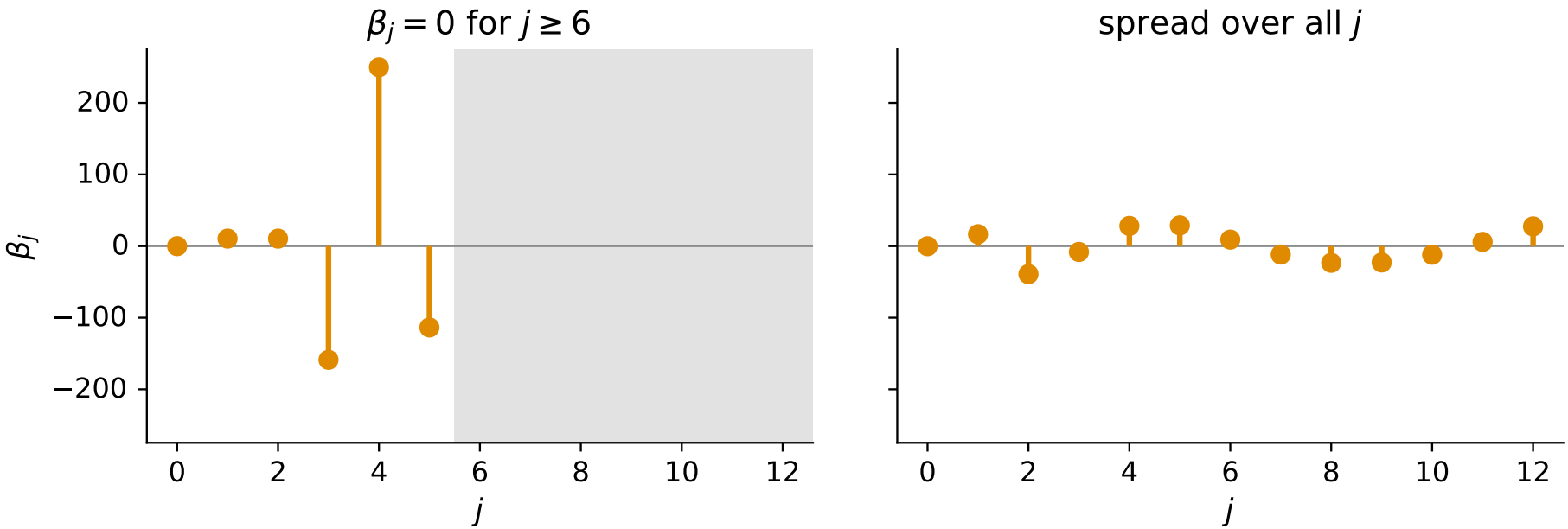
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1) When is having more, but smaller, coefficients preferable?
2) How do we implement this?

Optimization view of fitting a model

Consider the model

$$Y = \mu + \epsilon, \quad \epsilon \sim \text{Normal}(0, \sigma_\epsilon^2), \qquad N = N_{\text{train}}.$$

We could also write $\mu = \beta_0$ to use regression notation. When we fit this model to data, we are just estimating the mean, and we know that an unbiased estimator is

$$\hat{\mu} = \overline{Y} = \frac{1}{N} \sum_{i=1}^{N} Y_i.$$

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In the least-squares view, we interpret this estimator as the solution of an optimization problem. It minimizes

$$L(\mu) = \frac{\text{SSR}}{N} = \frac{1}{N} \sum_{i=1}^{N} (\mu - Y_i)^2 = \hat{\sigma}^2_{\text{train}},$$

where $L(\mu)$ is called the loss function.

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Regularization

The idea of regularization is to force μ to be smaller by add a penalty to the loss for larger μ:

$$L_\lambda(\mu) = \frac{1}{N}\sum_{i=1}^{N}(\mu - Y_i)^2 + \lambda\mu^2$$

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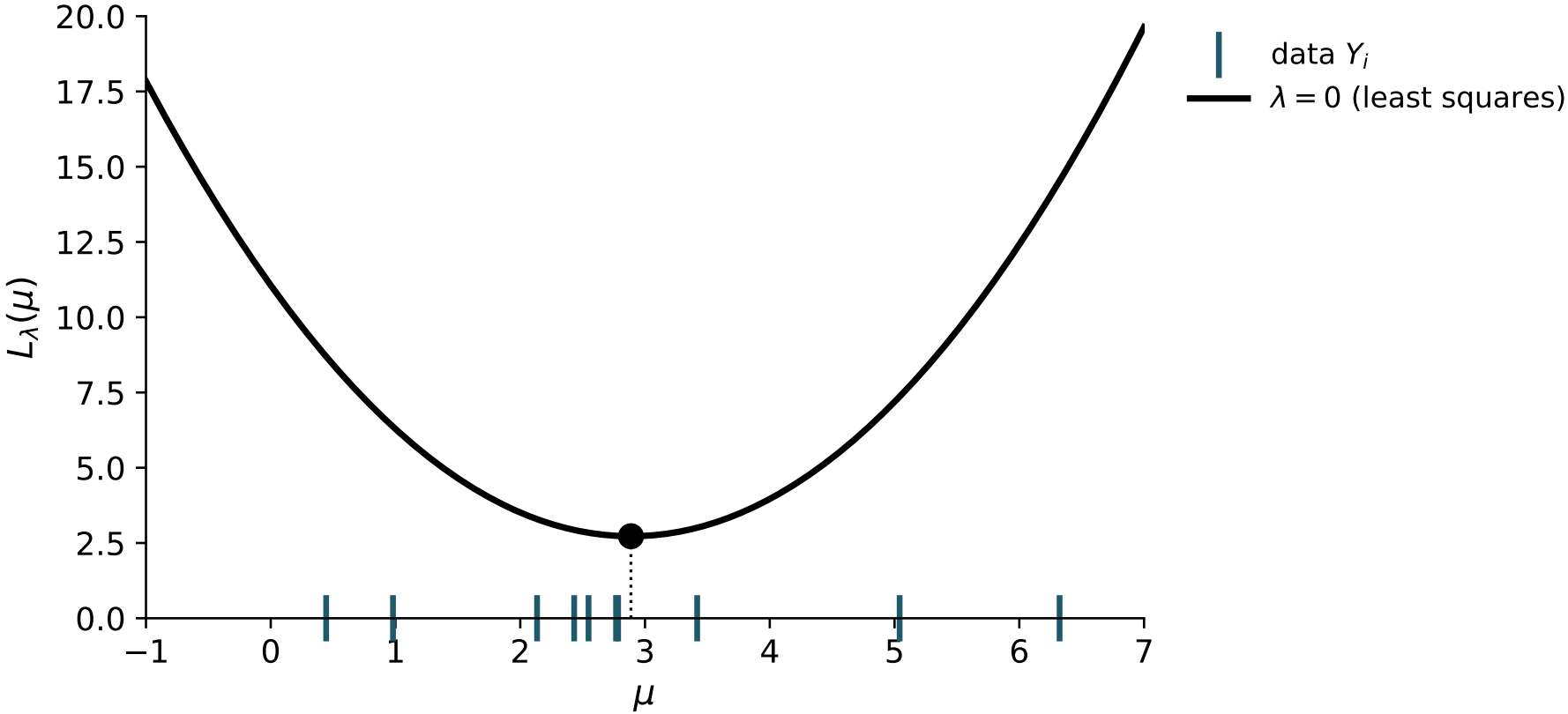
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As $\lambda \to \infty$ the estimate goes to 0. The larger λ, the smaller $\hat{\mu}^\lambda$.

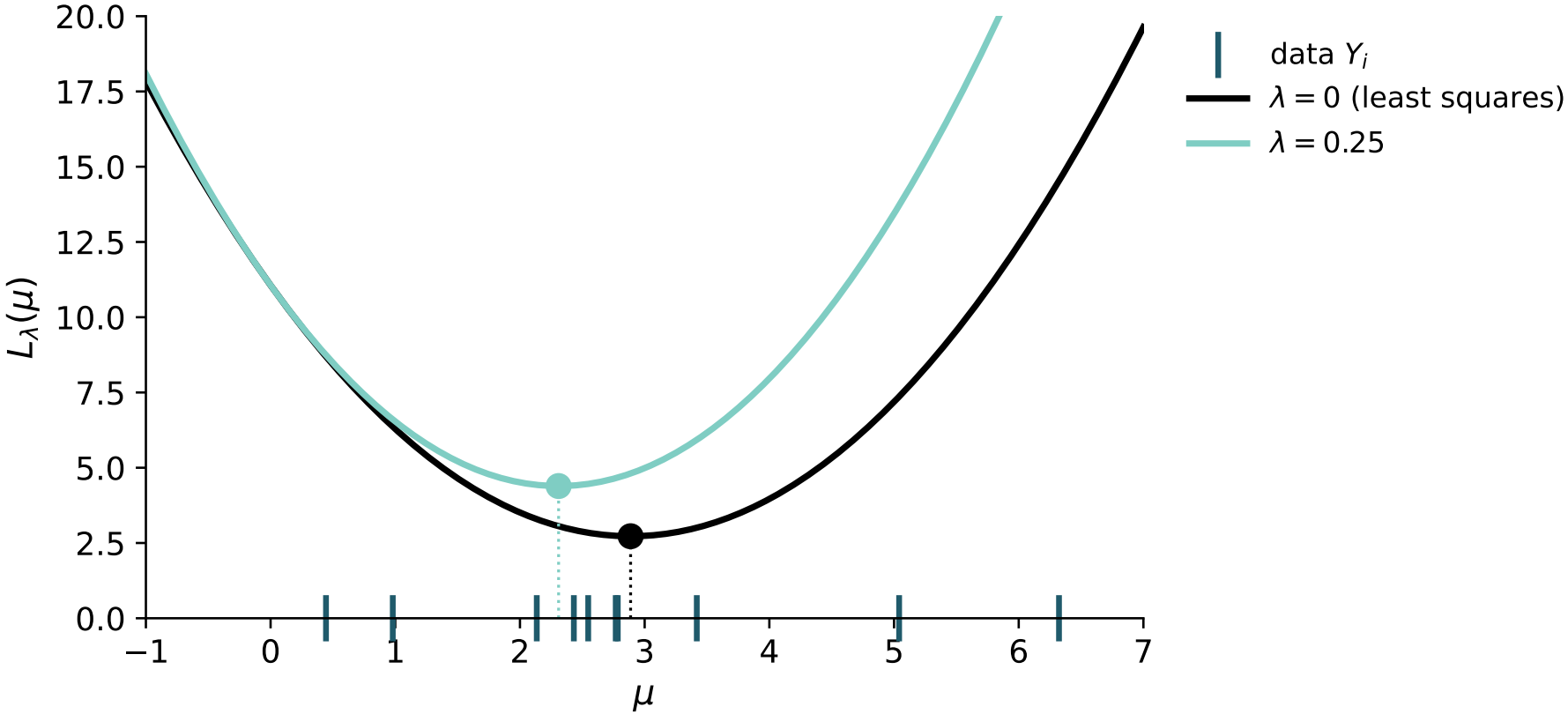
Loss for different λ



$$L_\lambda(\mu) = (1+\lambda)\left(\mu - \frac{\overline{Y}}{1+\lambda}\right)^2 + \hat{\sigma}_Y^2 + \frac{\lambda\,\overline{Y}^2}{1+\lambda}$$

Derive this: drill "Completing the square for the ridge loss of a mean", unit 6 notes §6.3.

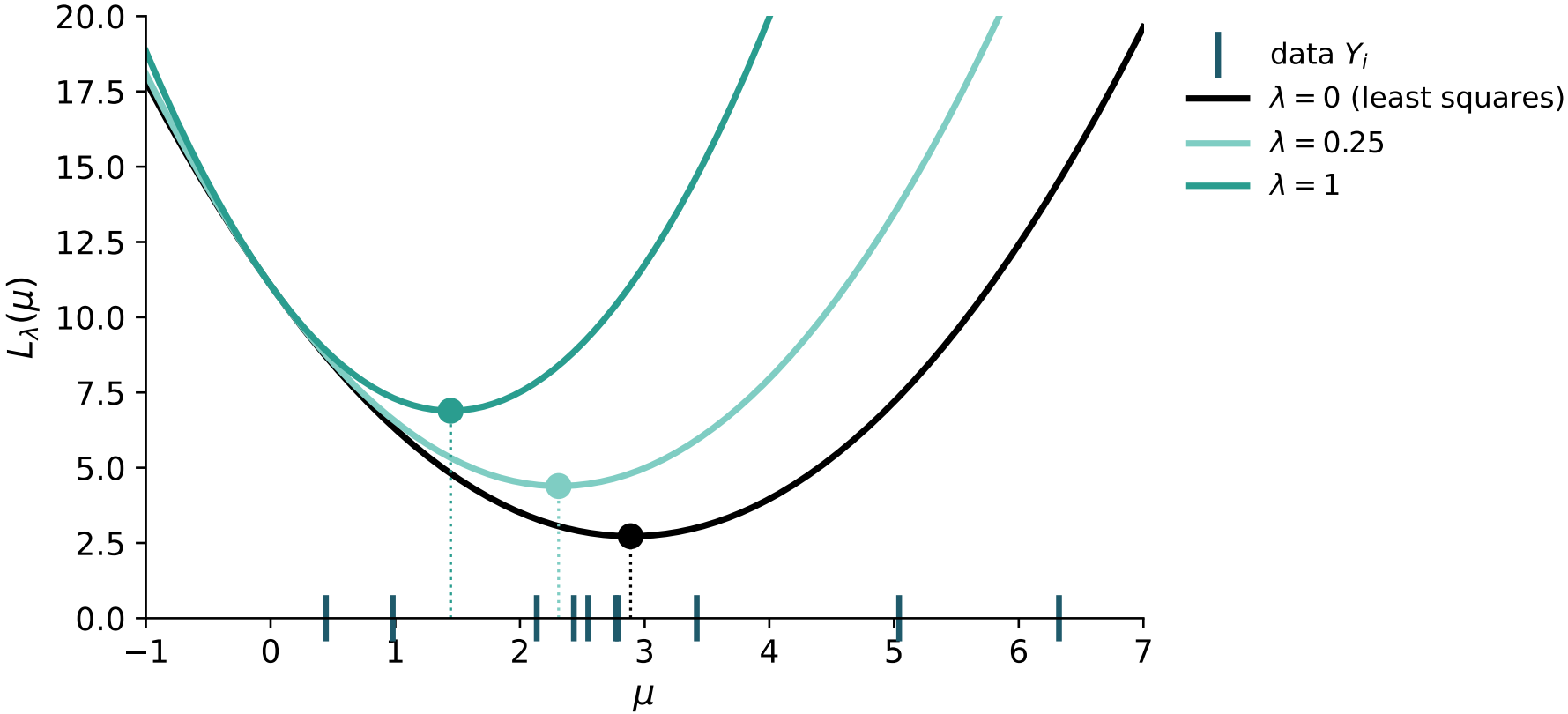
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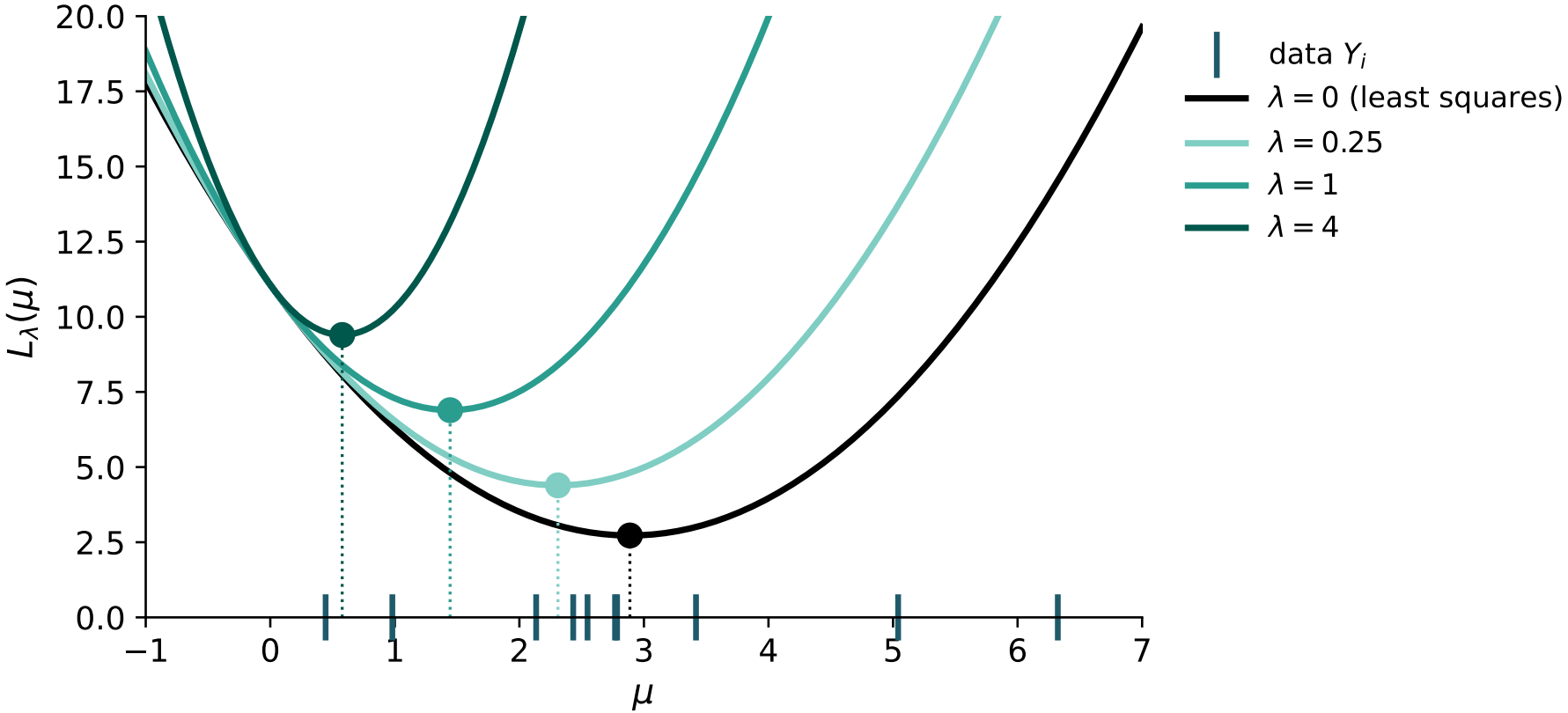
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Regularization for a linear model

Consider the model

$$Y = \beta_1 X + \epsilon, \quad \epsilon \sim \mathrm{Normal}(0, \sigma_\epsilon^2),$$

with $E[X] = 0$ and no intercept. The least-squares estimate is

$$\hat{\beta}_1 = \frac{\sum_i X_i Y_i}{\sum_i X_i^2}, \qquad \text{and we write} \quad \hat{\sigma}_X^2 = \frac{1}{N} \sum_i X_i^2.$$

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As $\lambda \to \infty$ the estimate goes to 0. The more spread out the X_i (larger $\hat{\sigma}_X^2$), the less a given λ shrinks it.

Regularization for a general regression model

least squares, degree K

$\hat{\beta}$ minimizes $\quad L(\beta) = \hat{\sigma}^2_{\text{train}} = \dfrac{1}{N_{\text{train}}} \displaystyle\sum_{i=1}^{N_{\text{train}}} \left(\hat{y}(X_i^{\text{train}}, D^{\text{train}}) - Y_i^{\text{train}}\right)^2$

each x^j is either in the model ($j \leq K$) or not

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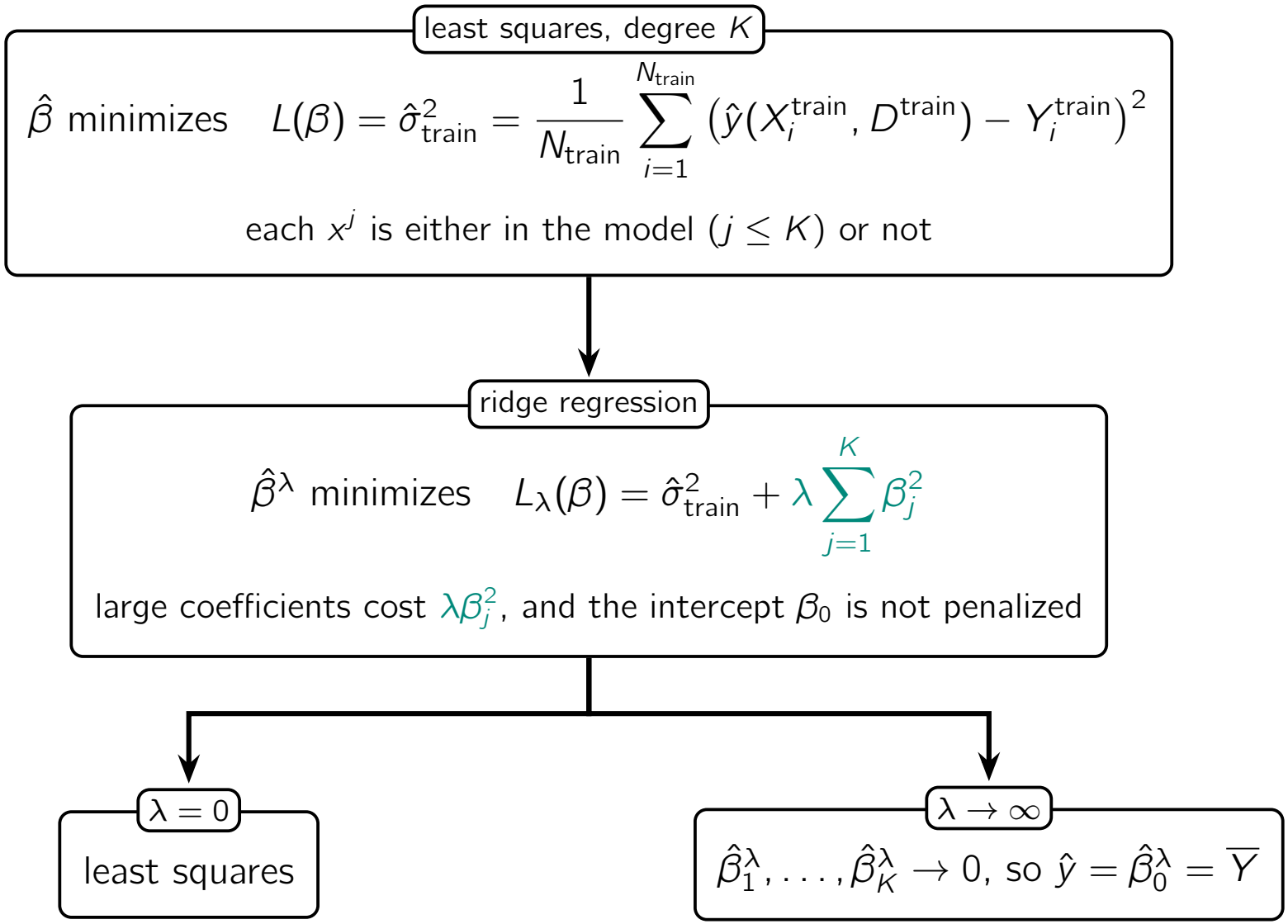
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ridge regression

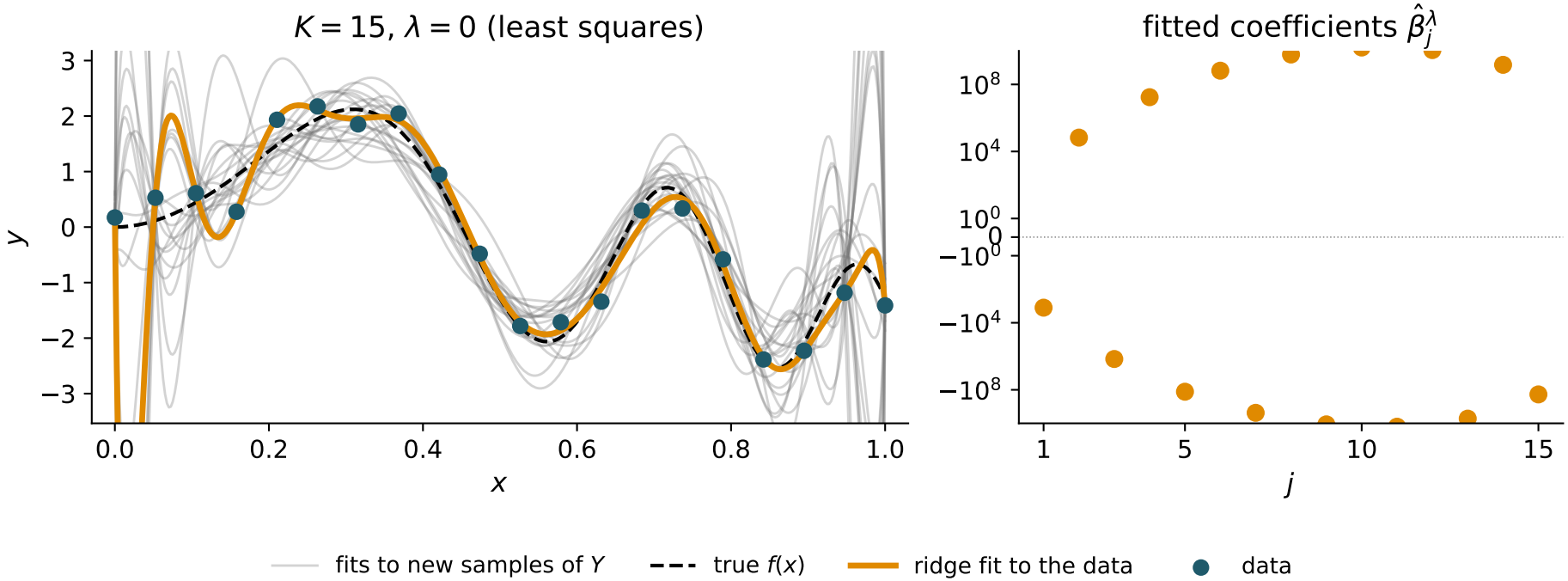
$\hat{\beta}^\lambda$ minimizes $\quad L_\lambda(\beta) = \hat{\sigma}^2_{\text{train}} + \lambda \displaystyle\sum_{j=1}^{K} \beta_j^2$

large coefficients cost $\lambda\beta_j^2$, and the intercept β_0 is not penalized

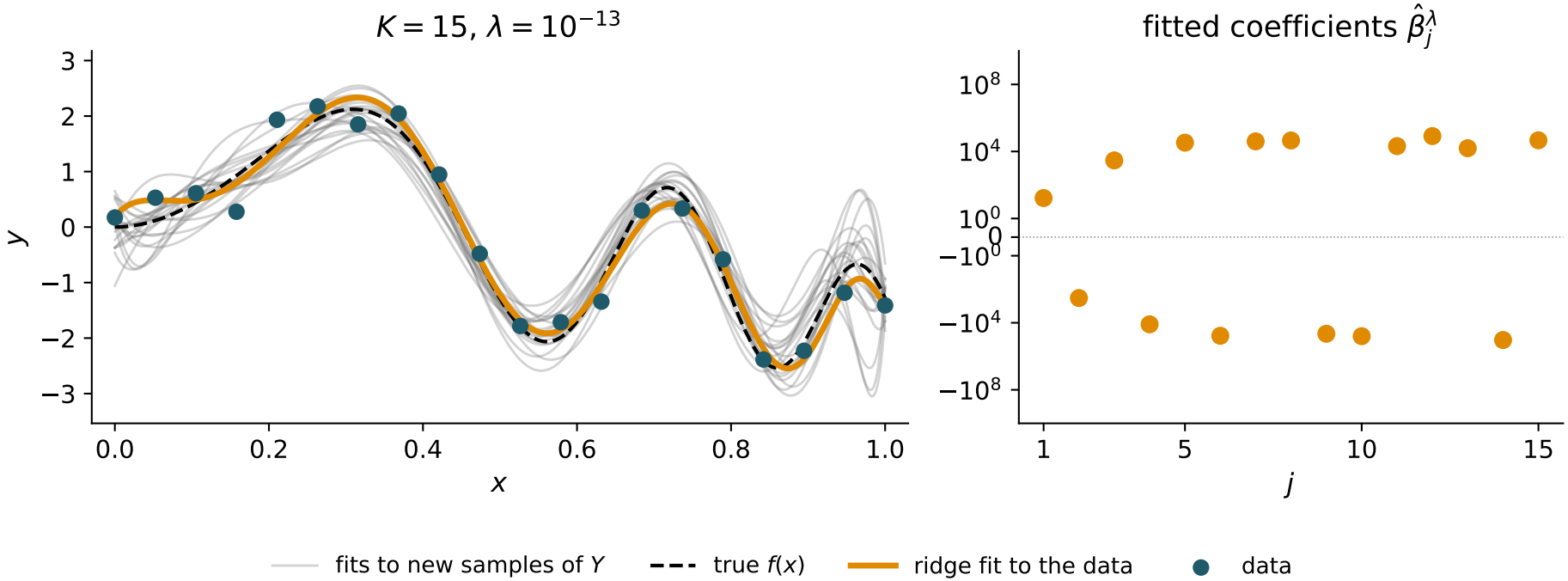
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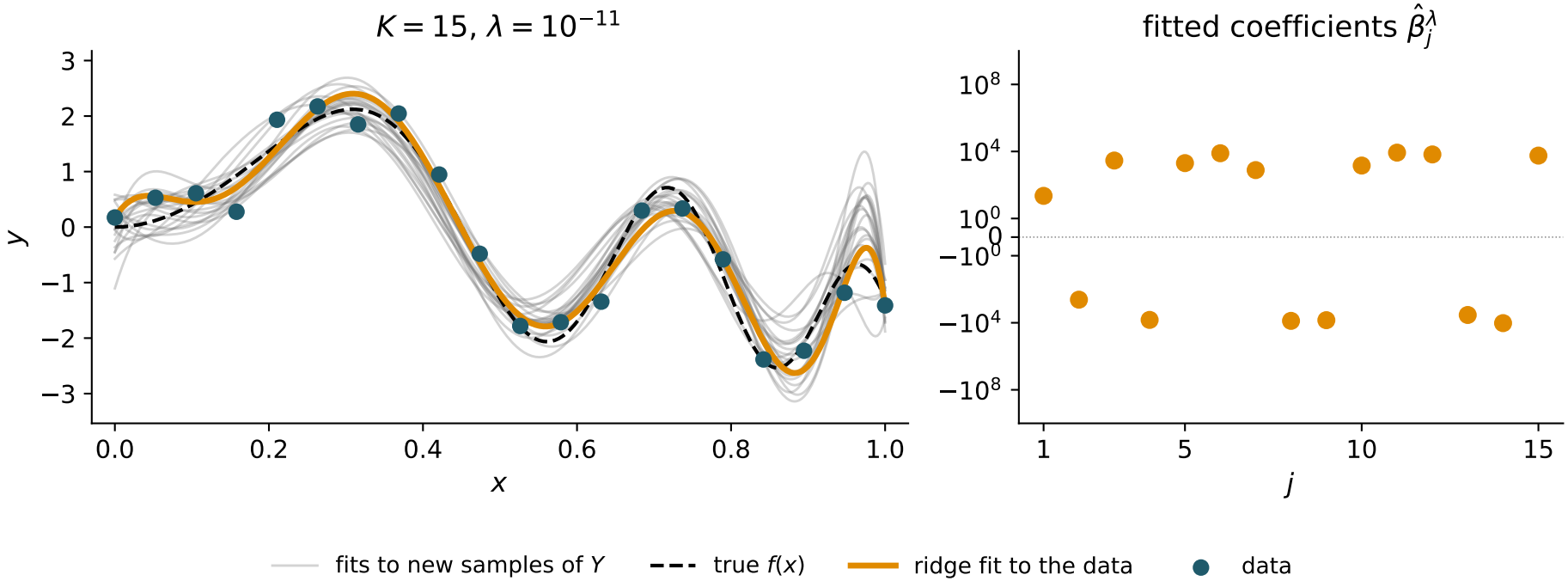
The effect of λ



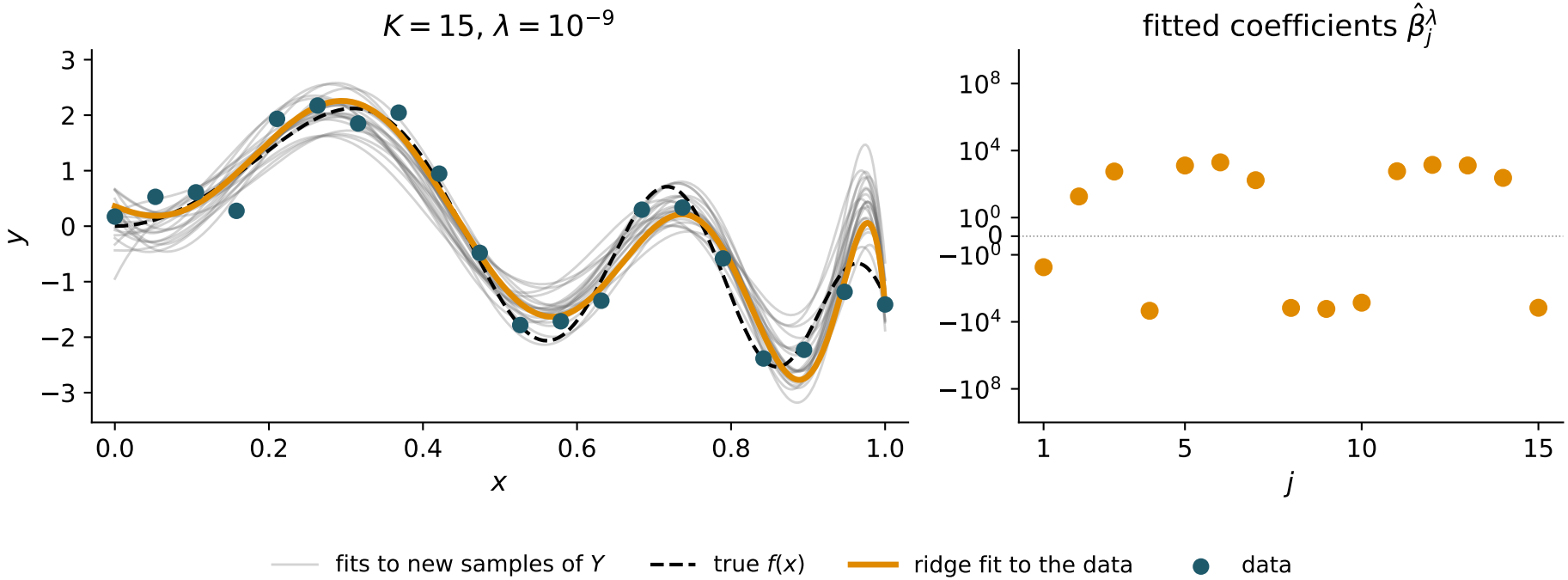
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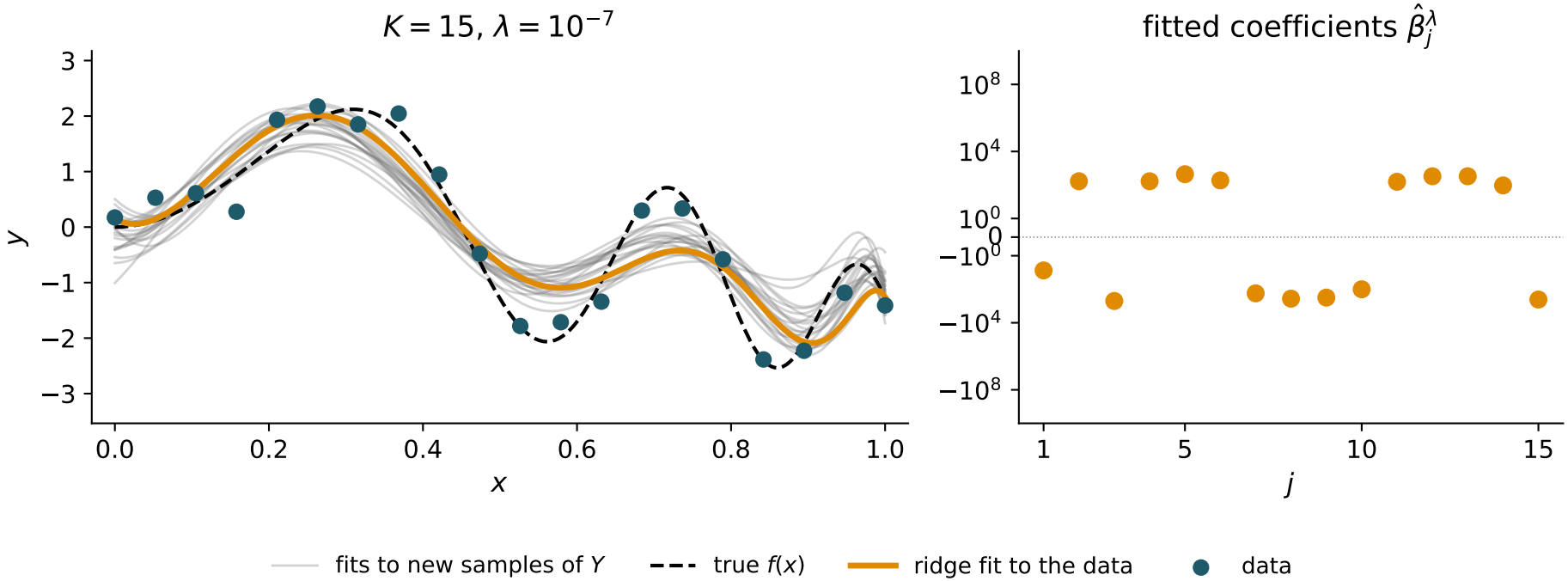
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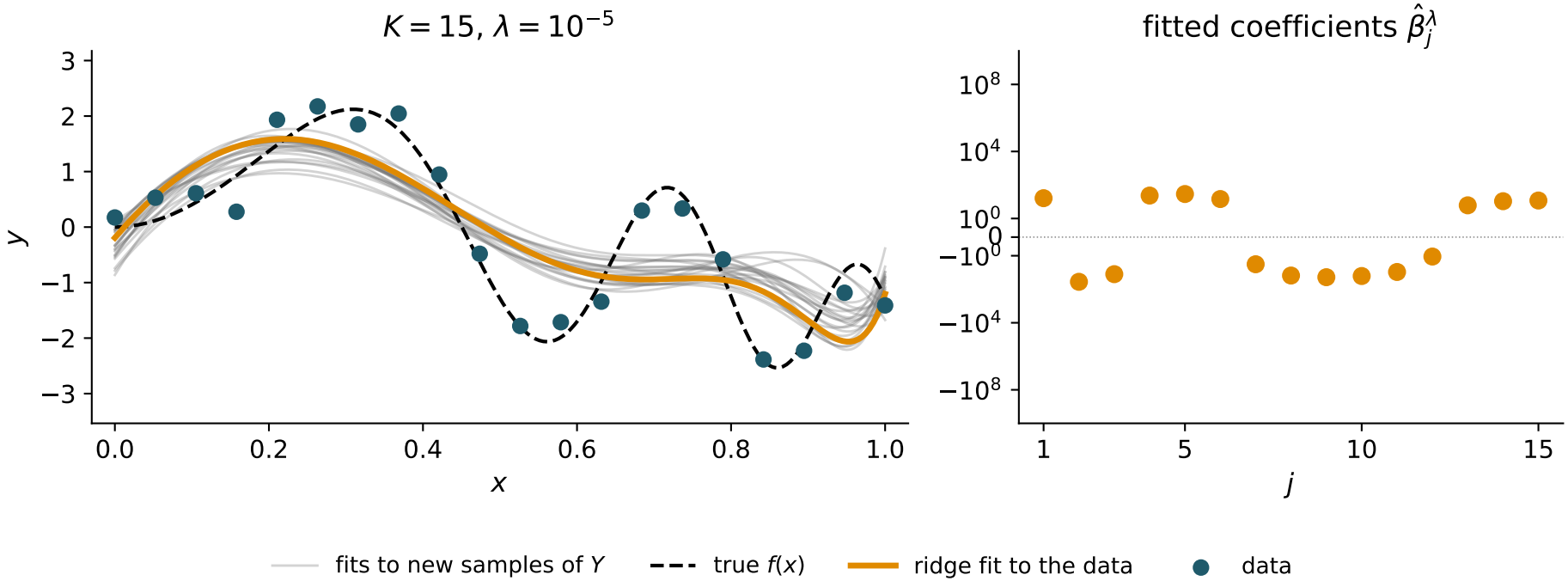
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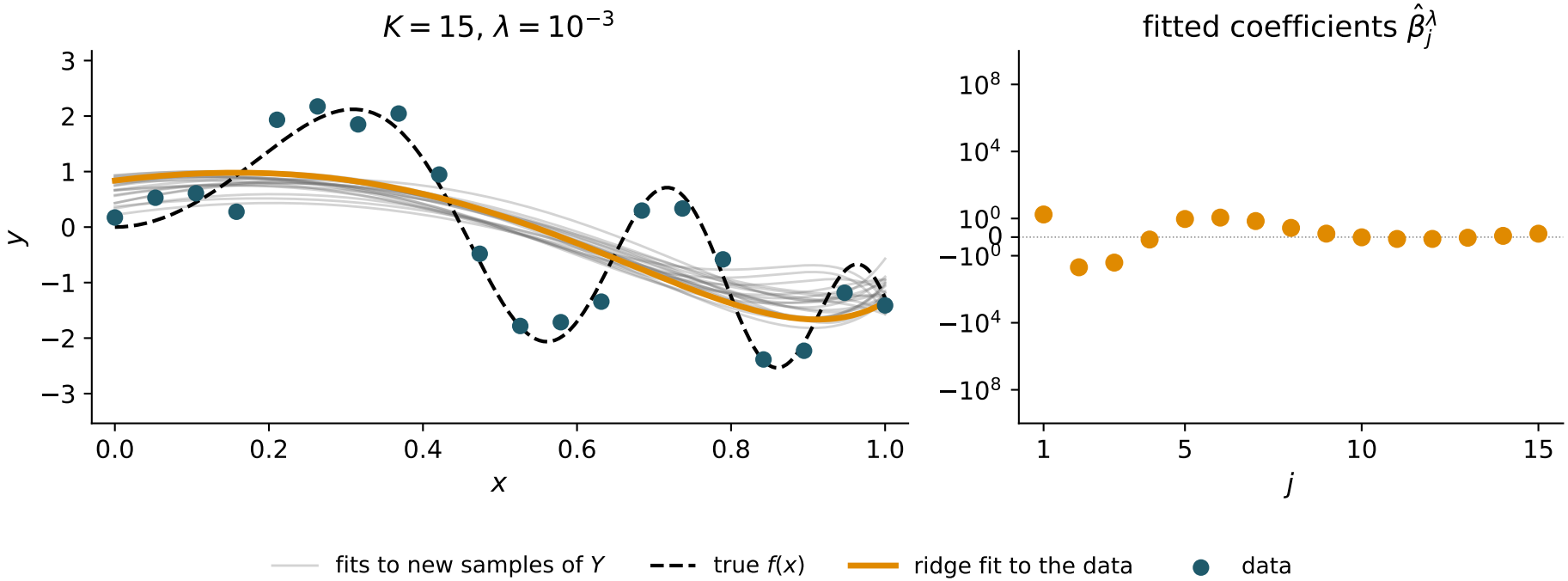
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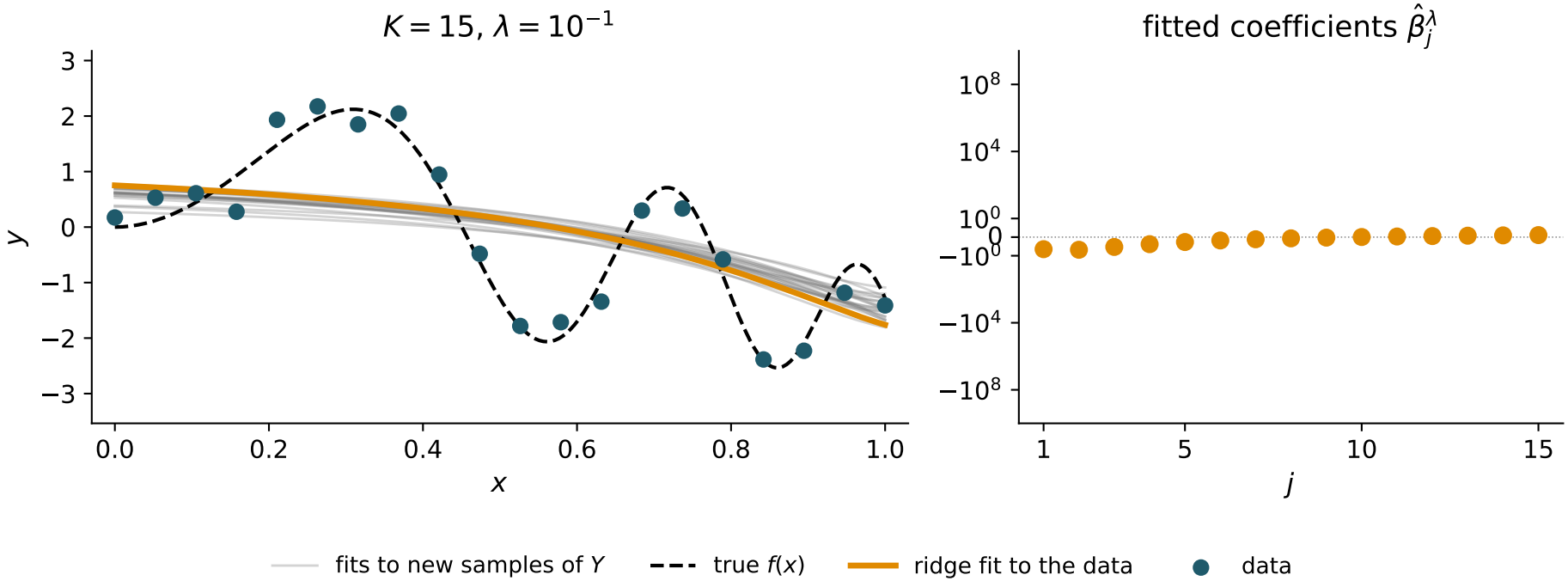
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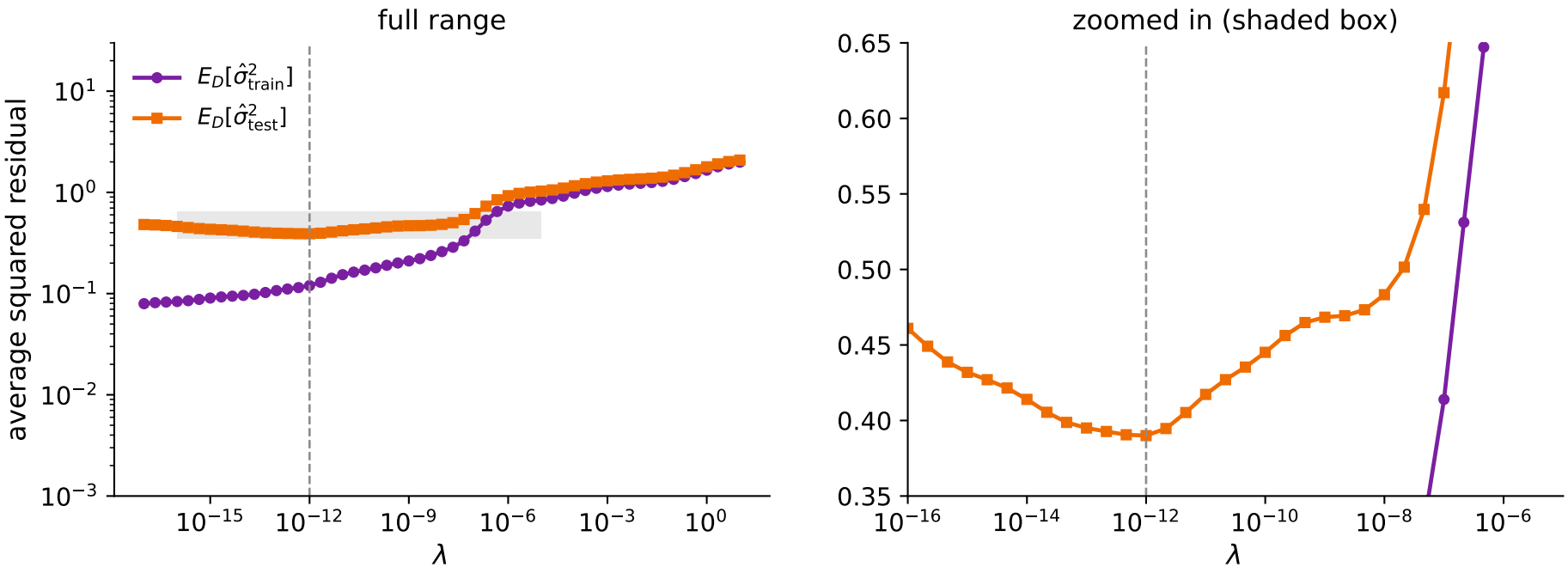


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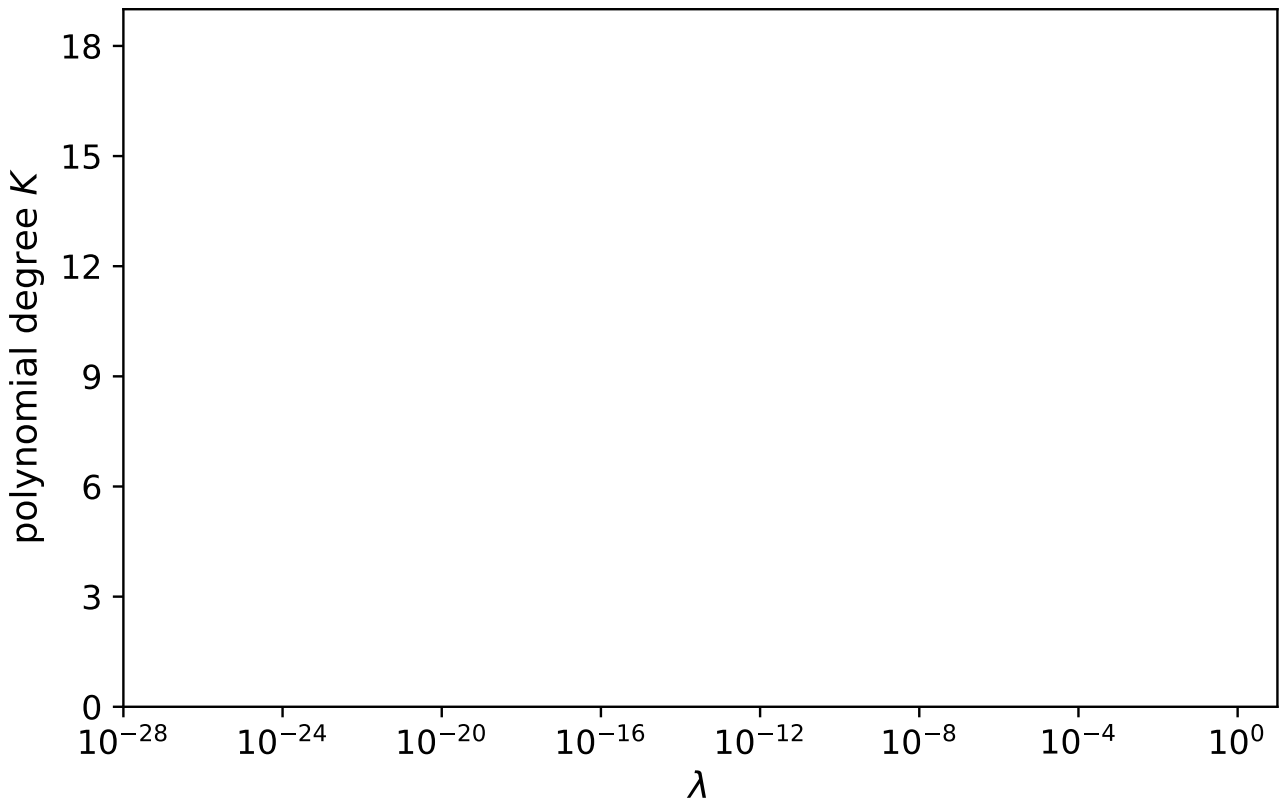
Training and test error vs. λ

Each error is averaged over 400 simulated training sets of the same 20 x values.



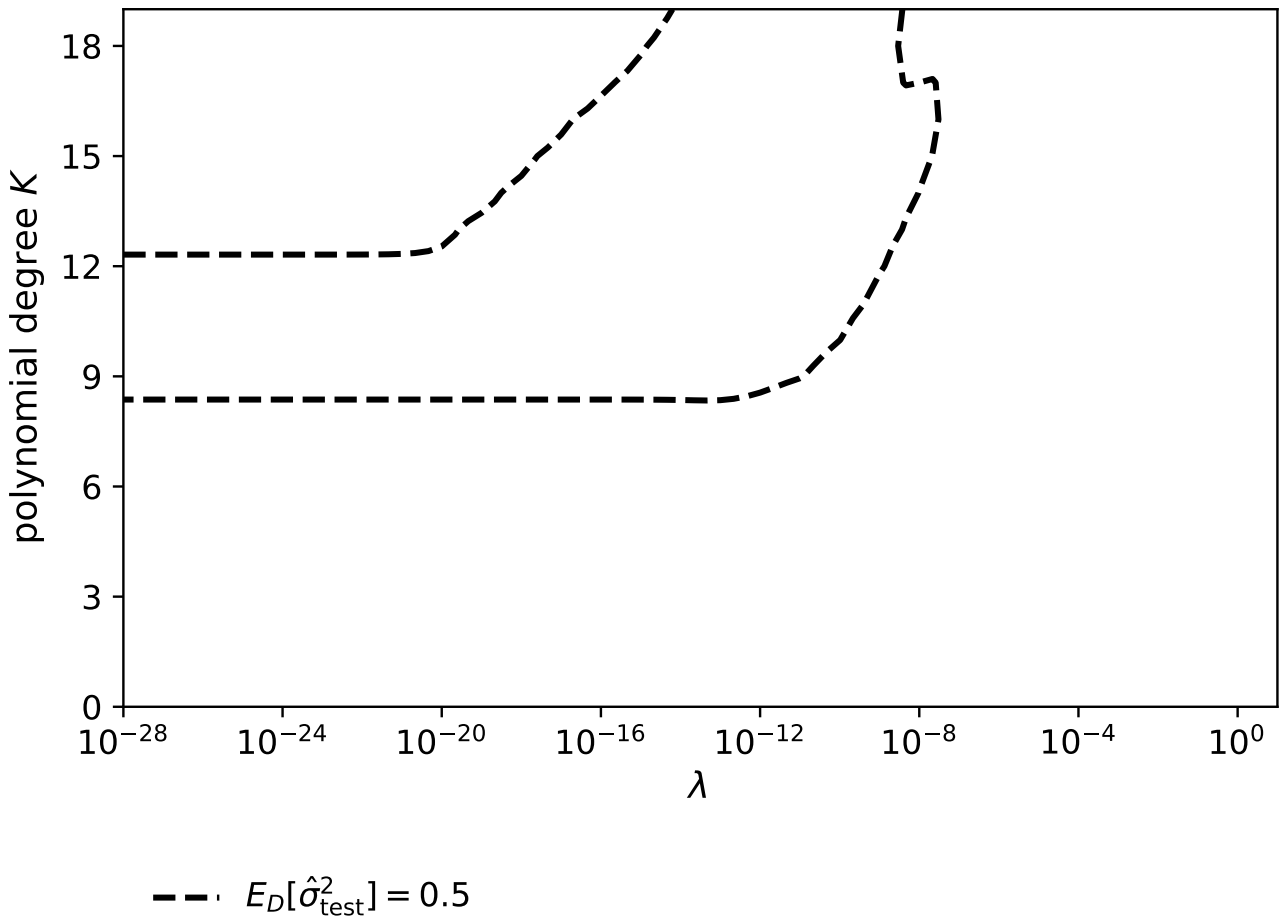
Understanding the error in the K–λ plane

Each point is a choice of degree K and penalty λ. What do the lines where $E_D[\hat{\sigma}^2_{\text{test}}]$ is constant look like?

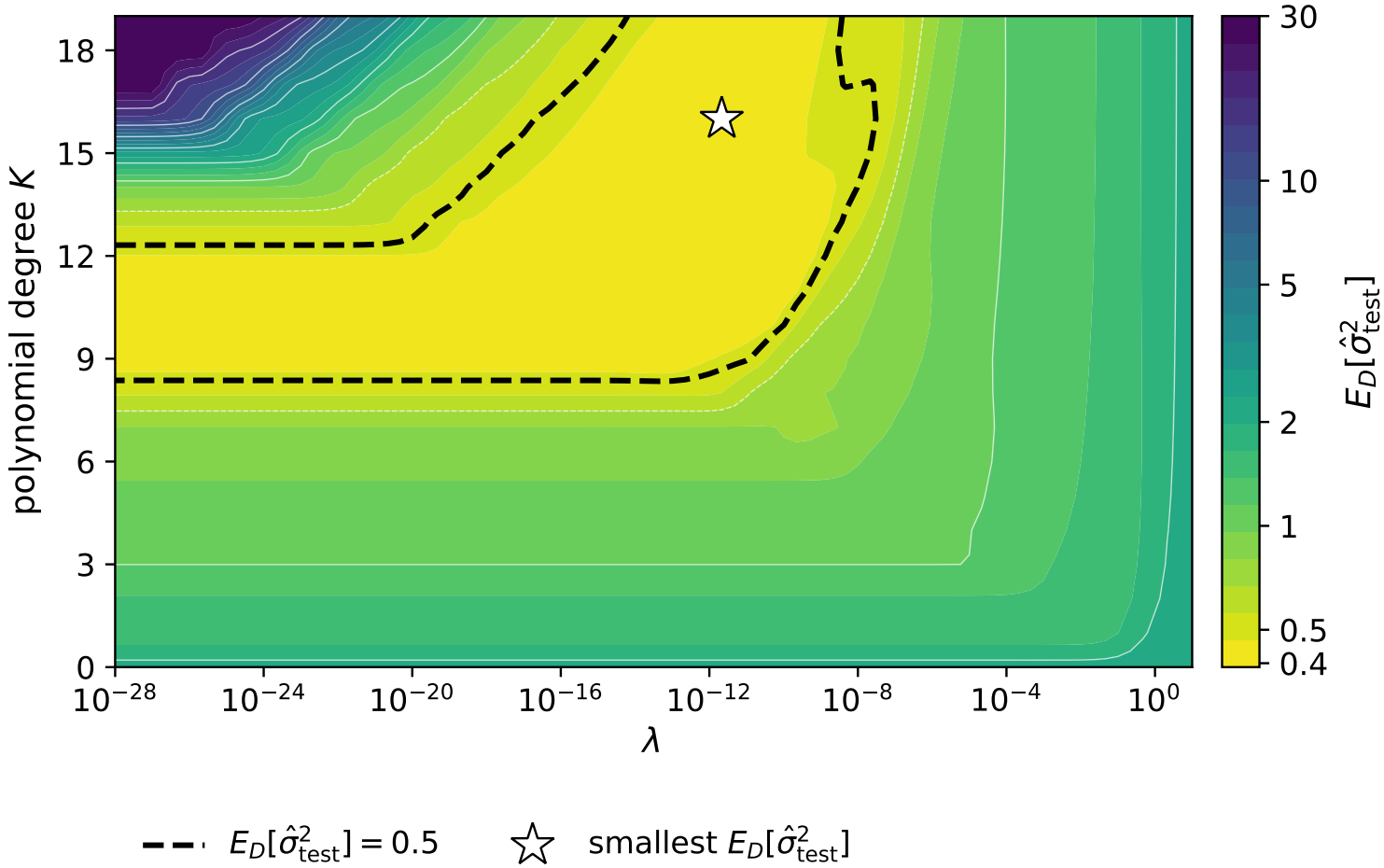


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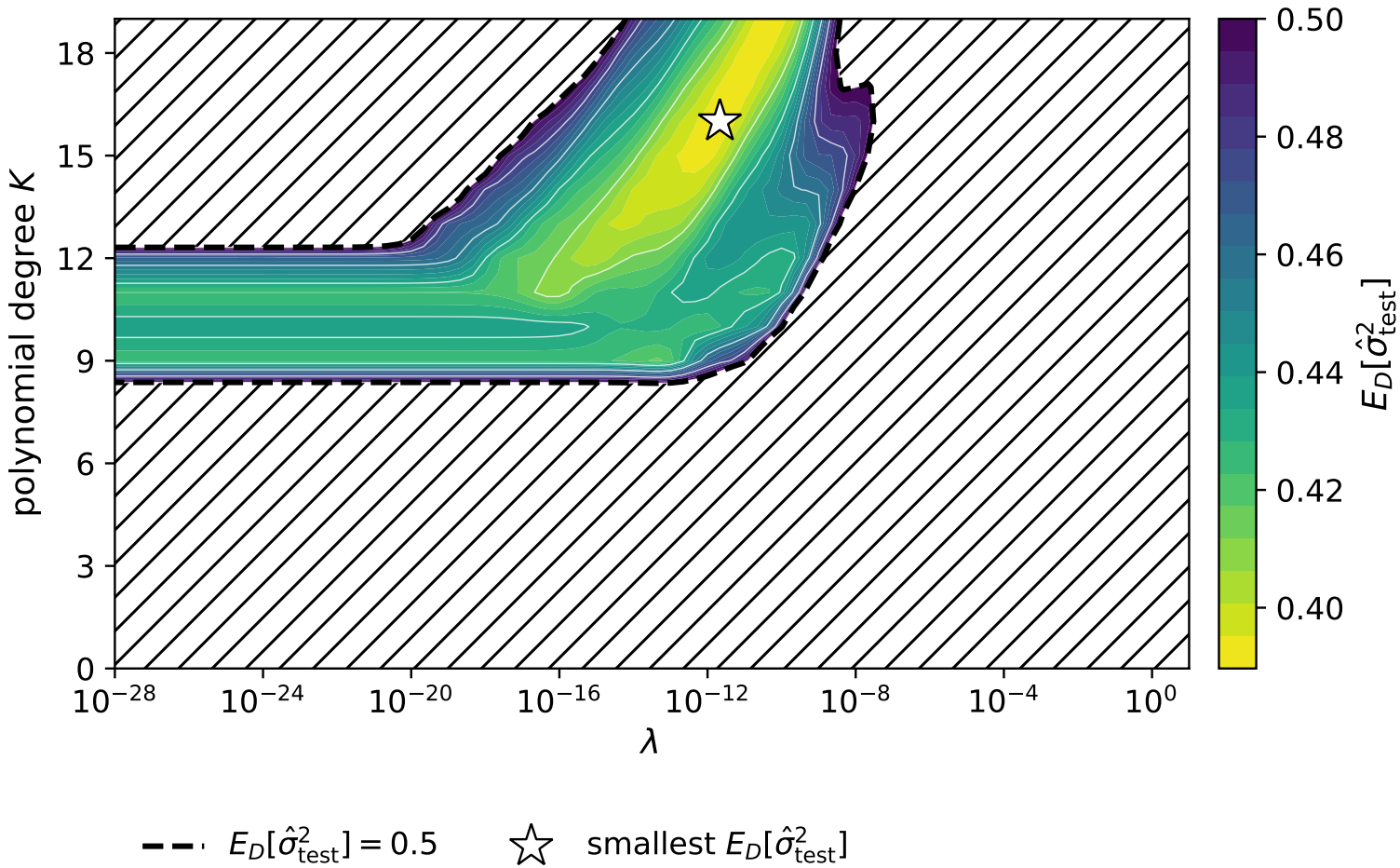
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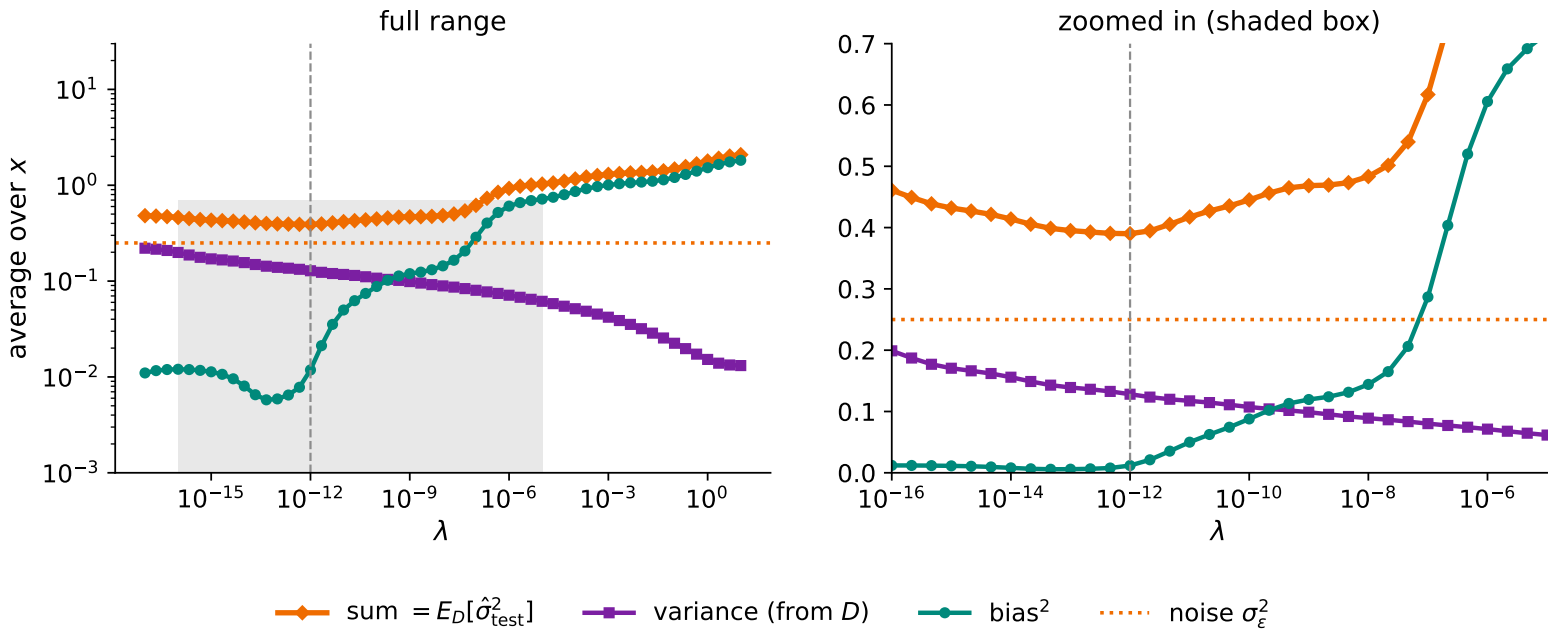
Test error over K and λ



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Bias and variance vs. λ



The spike protein model with regularization

probability model: interactions up to order 5

$$Y = \beta_0 + \sum_i \beta_i X_i + \sum_{i<k} J_{ik} X_i X_k + \sum_{i<k<l} J_{ikl} X_i X_k X_l + \cdots + \epsilon$$

$X_i = 1$ if mutation i is present, $i = 1, \ldots, 15$, and 4944 coefficients in all

The spike protein model with regularization

probability model: interactions up to order 5

$$Y = \beta_0 + \sum_i \beta_i X_i + \sum_{i<k} J_{ik} X_i X_k + \sum_{i<k<l} J_{ikl} X_i X_k X_l + \cdots + \epsilon$$

$X_i = 1$ if mutation i is present, $i = 1, \ldots, 15$, and 4944 coefficients in all

training set

$N_{\text{train}} = 4000$ variants

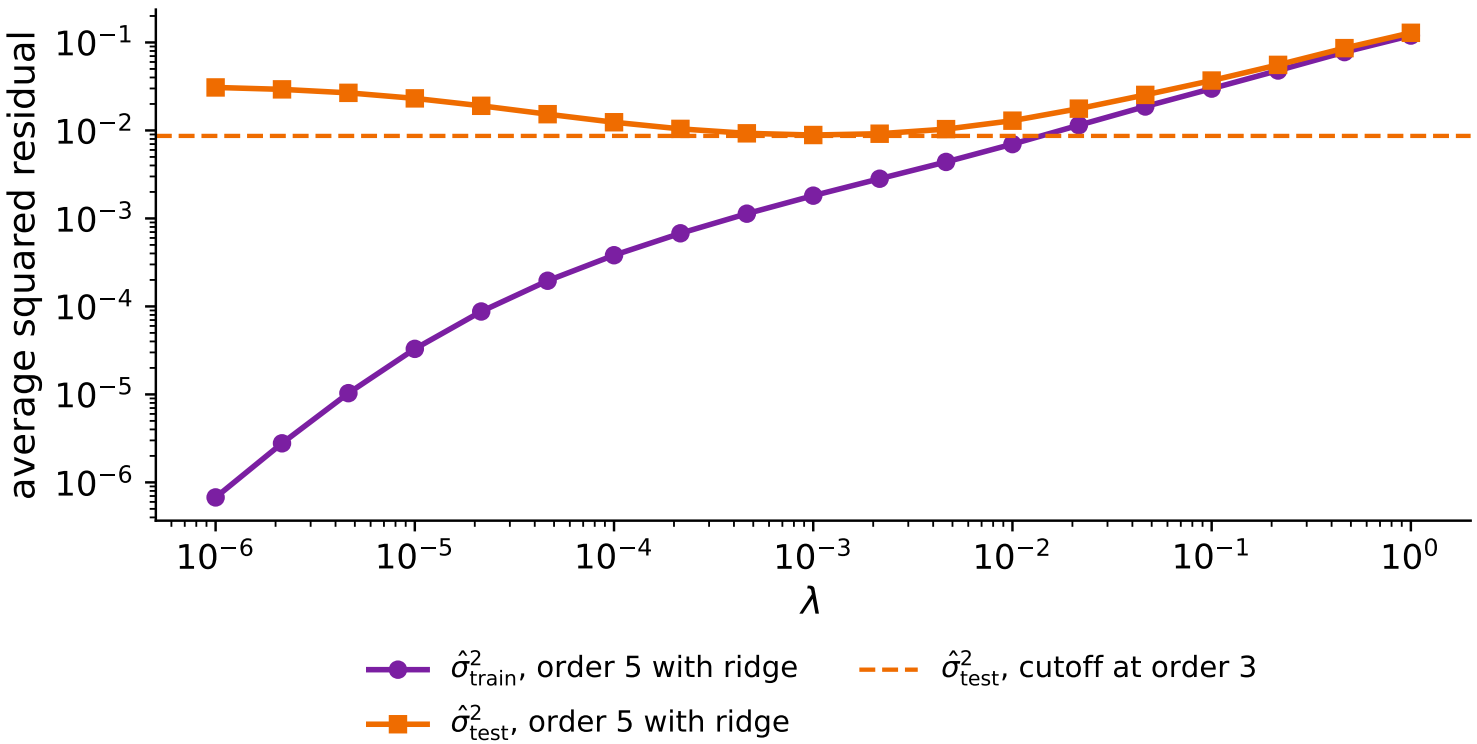
fit with ridge regression

minimize $\quad L_\lambda = \hat{\sigma}^2_{\text{train}} + \lambda\left(\sum_i \beta_i^2 + \sum_{i<k} J_{ik}^2 + \sum_{i<k<l} J_{ikl}^2 + \cdots\right)$

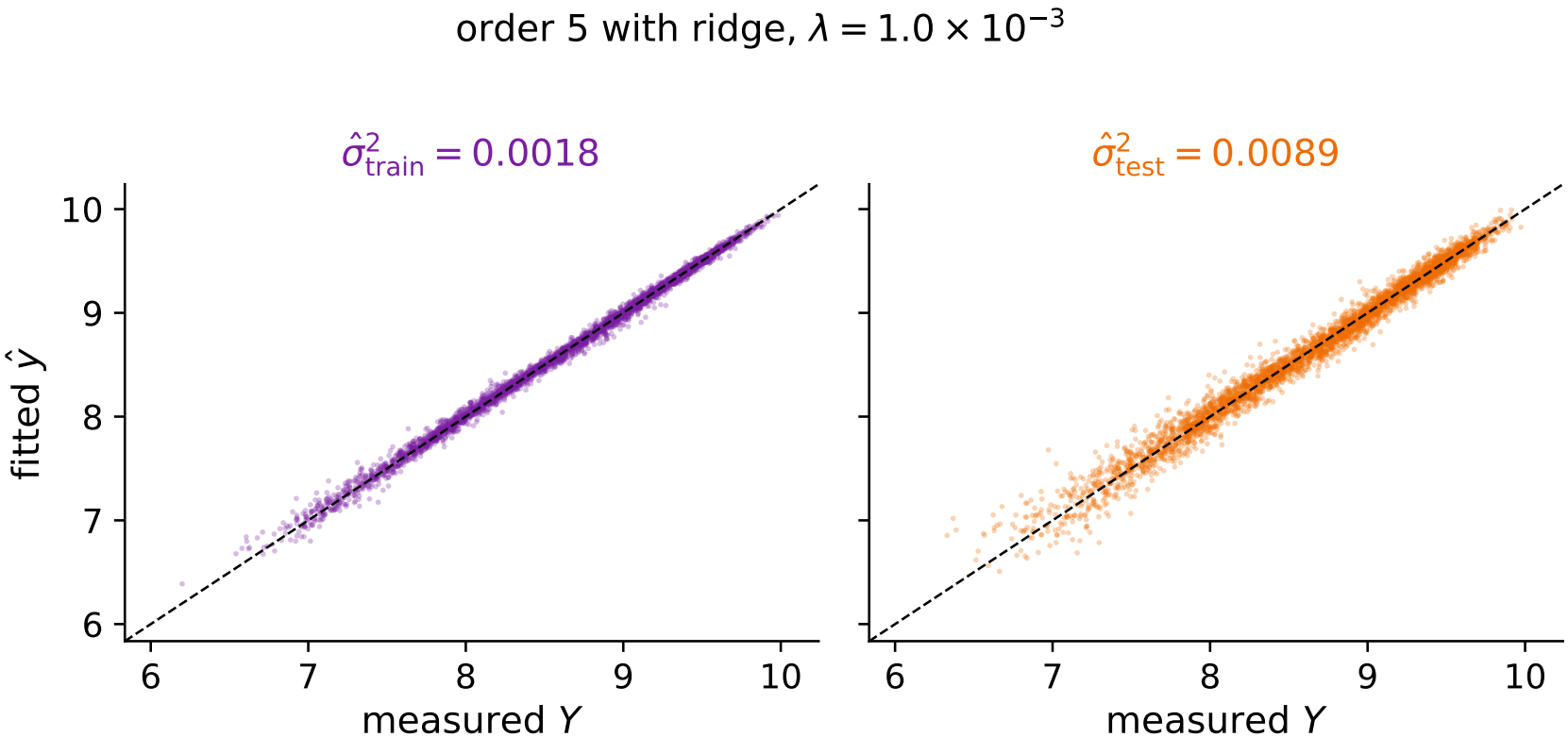
every coefficient except β_0 gets the same penalty

Regularization for the spike protein

All interactions up to order 5 (4944 coefficients), fit with ridge to the same 4000 training variants as on 11/6 and scored on the same 4000 test variants. The dashed line is least squares with the cutoff at order 3.



Fitted vs. measured binding with ridge



Regularization for images: setup

data: a noisy satellite image



$N = 160 \times 160$ pixels

Y = brightness at pixel (x, y)

Regularization for images: setup

data: a noisy satellite image



$N = 160 \times 160$ pixels

Y = brightness at pixel (x, y)

representation in feature space

$$\phi_{k,l}(x, y) = \cos(\pi k x)\cos(\pi l y)$$

$k, l = 0, \ldots, 159$: one feature for each pixel

Regularization for images: setup

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probability model

$$Y = \sum_{k,l} \beta_{k,l}\, \phi_{k,l}(x, y) + \epsilon$$

Regularization for images: setup

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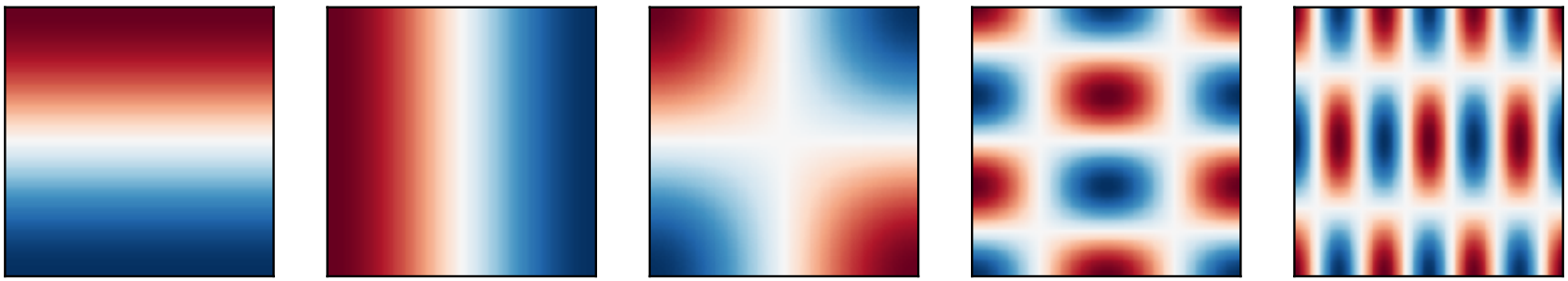
regularization

$$L_\lambda(\beta) = \hat{\sigma}^2_{\text{train}} + \lambda \sum_{k,l} (k^2 + l^2)\, \beta^2_{k,l}$$

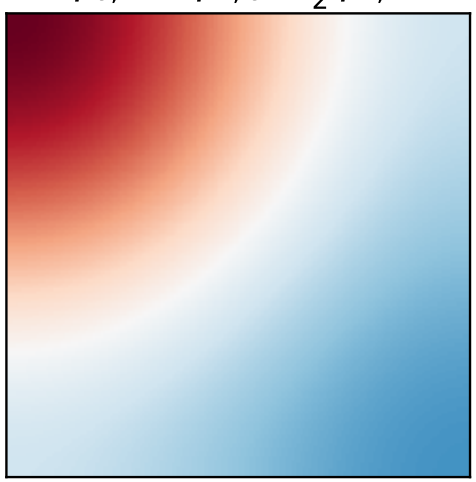
high frequencies cost more

The Fourier features

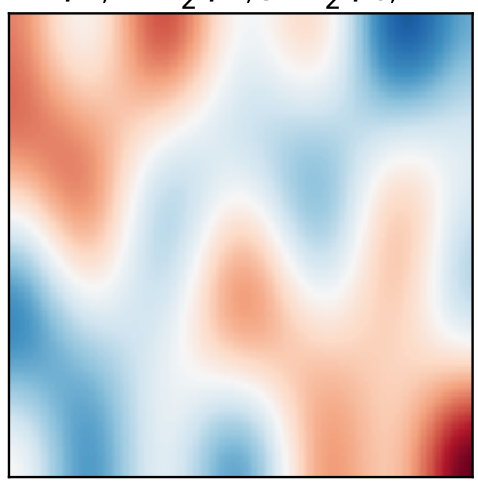
$\phi_{0,1}$ $\phi_{1,0}$ $\phi_{1,1}$ $\phi_{2,3}$ $\phi_{6,2}$



$\phi_{0,1} + \phi_{1,0} + \frac{1}{2}\phi_{1,1}$

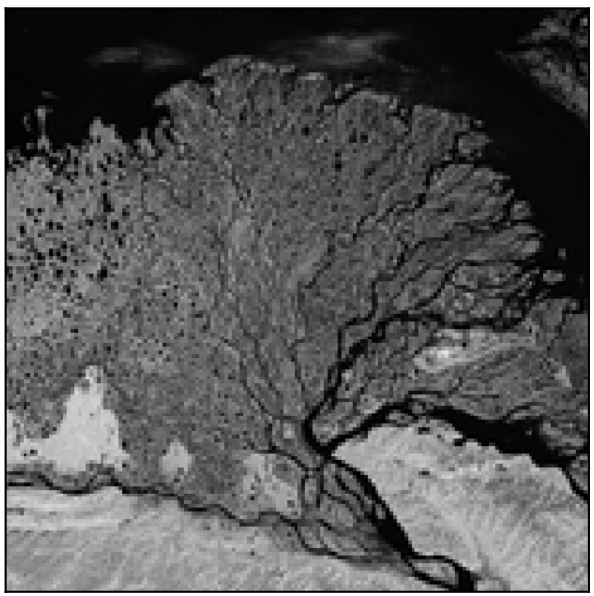


$\phi_{1,1} - \frac{1}{2}\phi_{2,3} + \frac{1}{2}\phi_{6,2}$

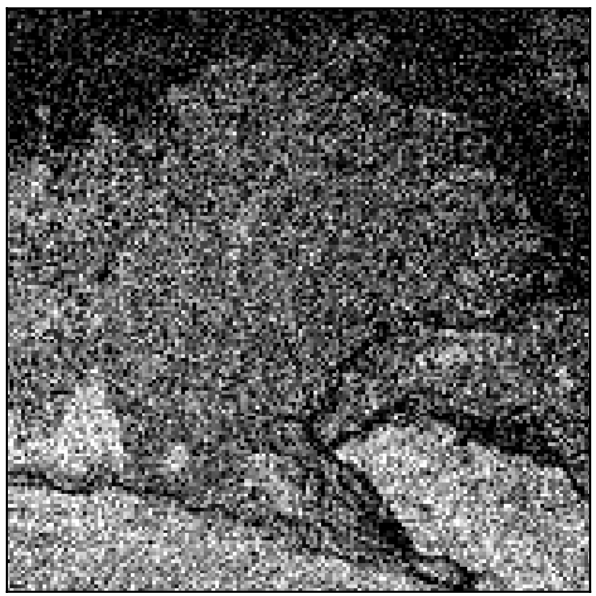


Smoothing a noisy image

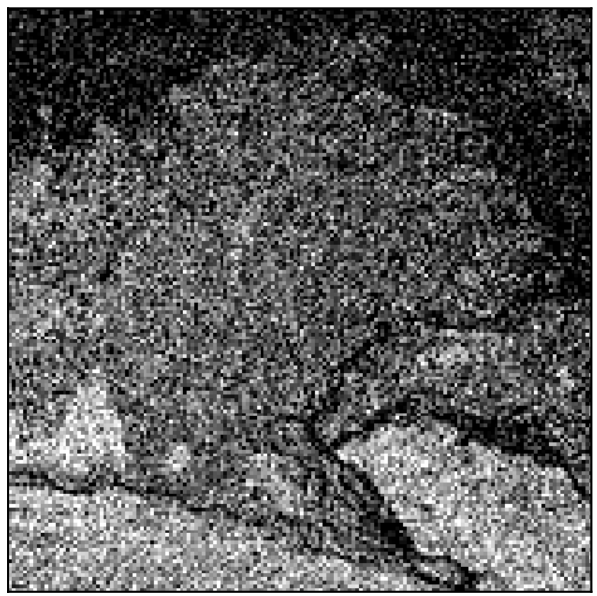
clean image (unknown)



noisy image (data)



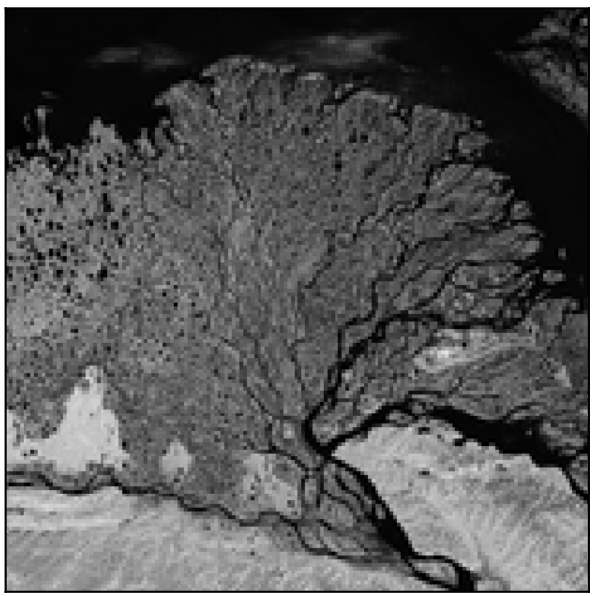
fit, $\lambda = 0$ (least squares)



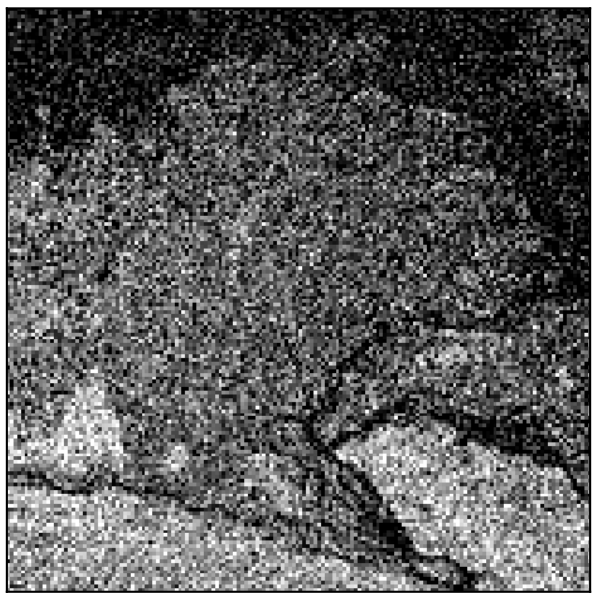
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

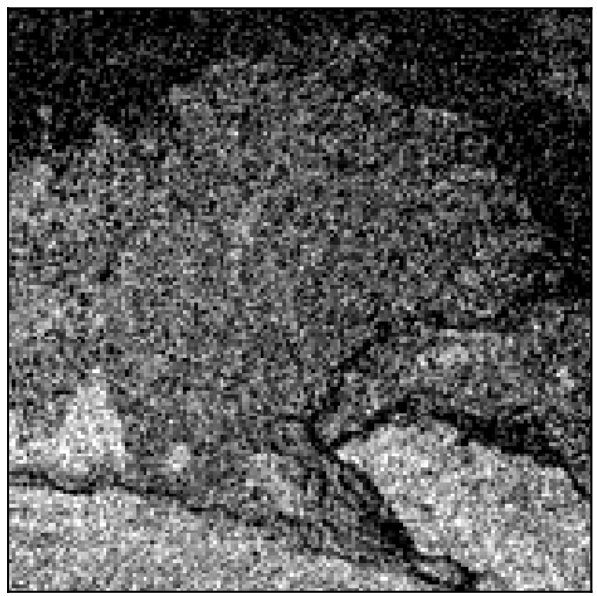
clean image (unknown)



noisy image (data)



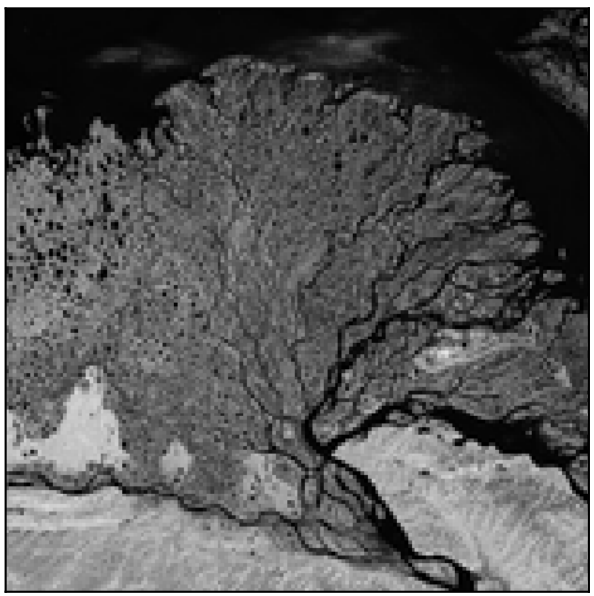
fit, $\lambda = 10^{-9}$



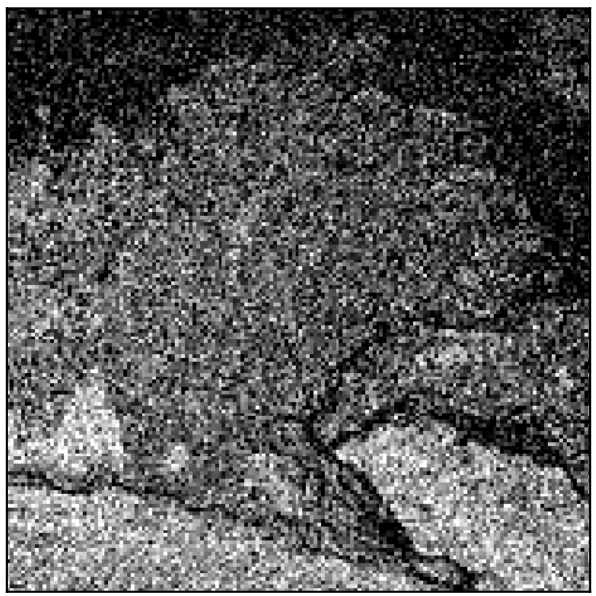
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

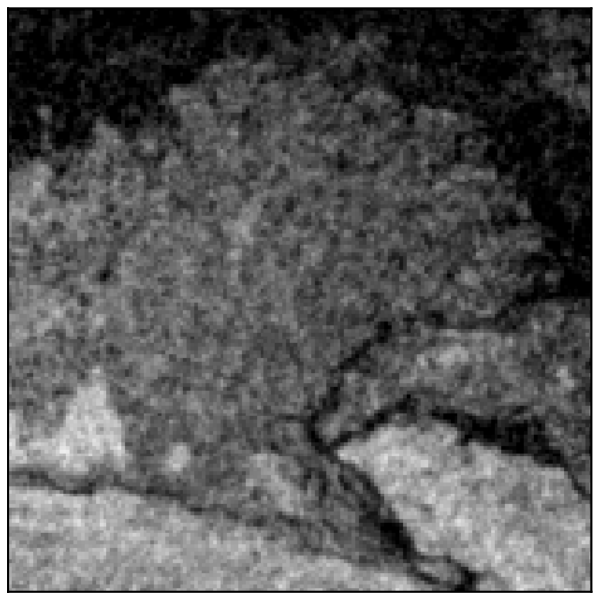
clean image (unknown)



noisy image (data)



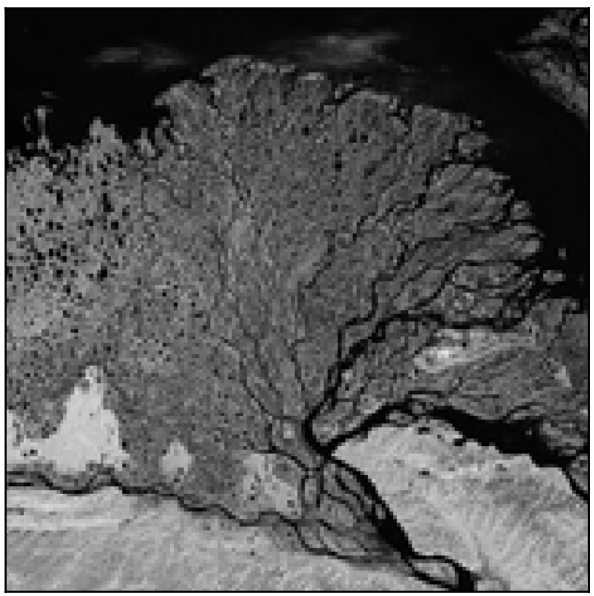
fit, $\lambda = 10^{-8}$



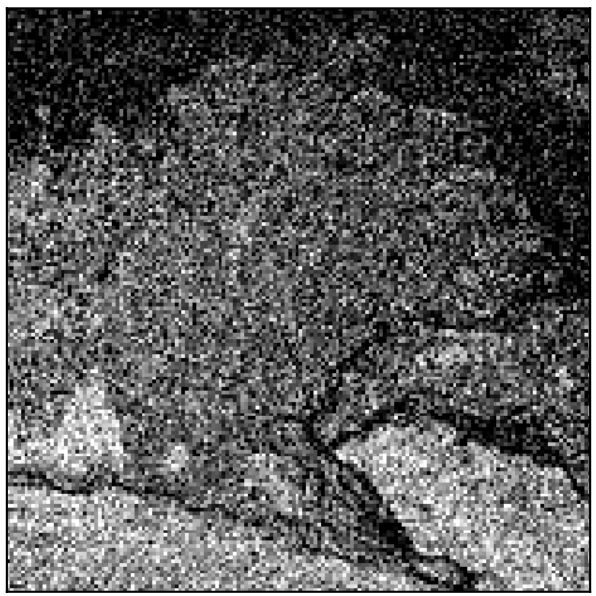
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

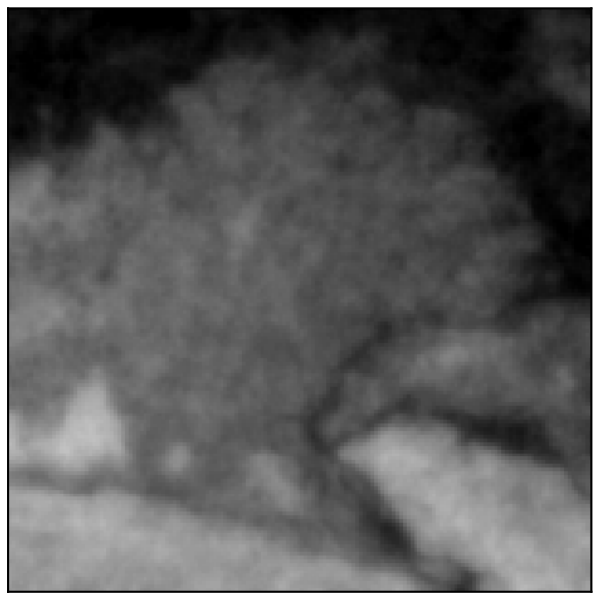
clean image (unknown)



noisy image (data)



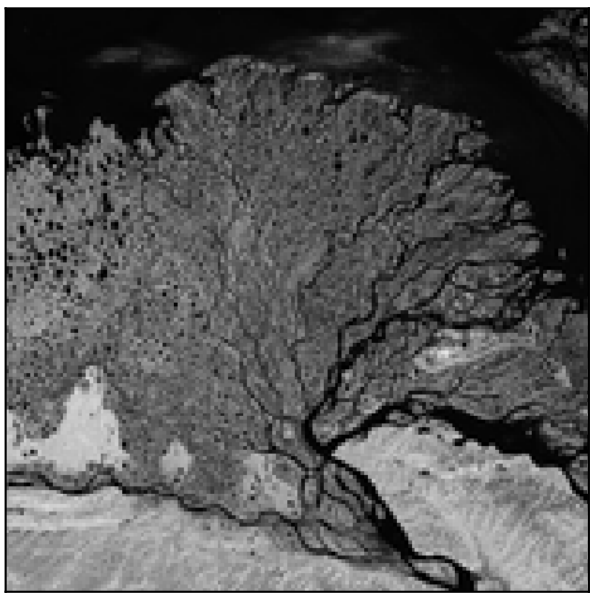
fit, $\lambda = 10^{-7}$



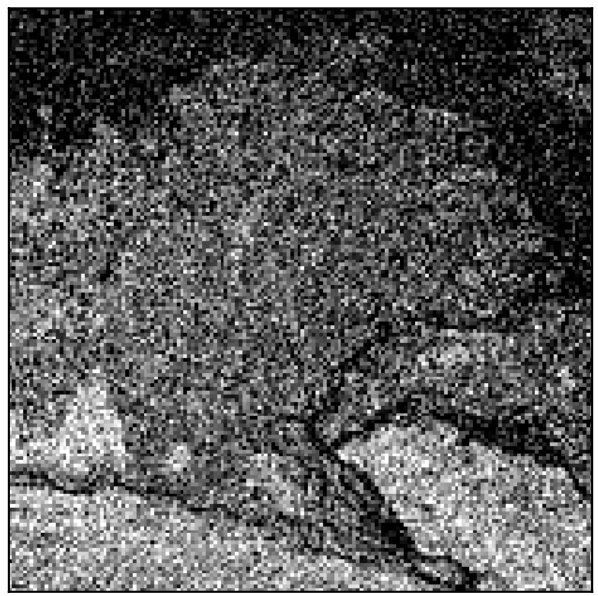
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

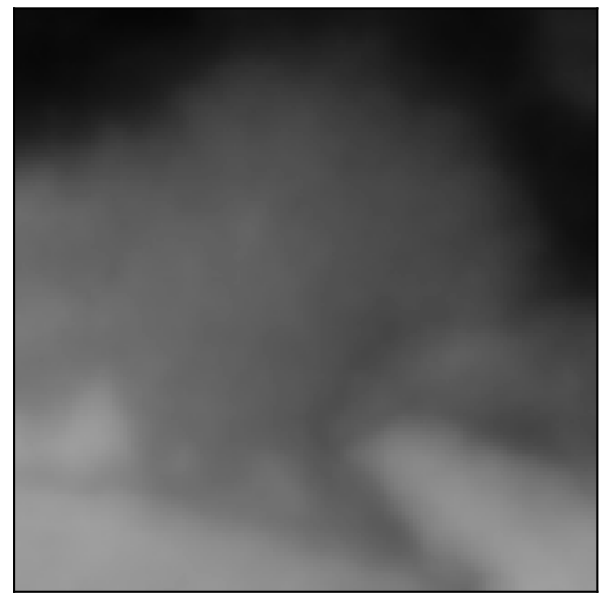
clean image (unknown)



noisy image (data)



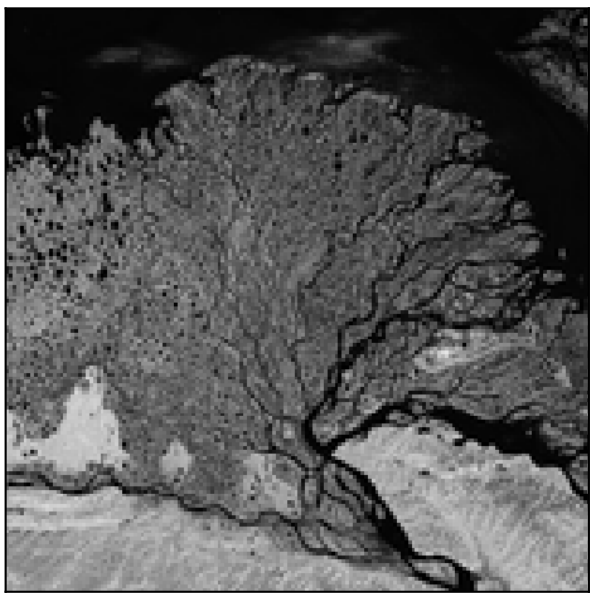
fit, $\lambda = 10^{-6}$



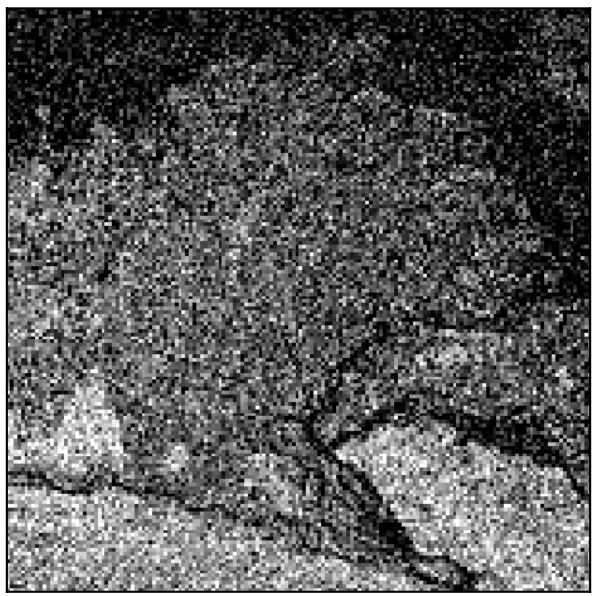
Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Smoothing a noisy image

clean image (unknown)



noisy image (data)

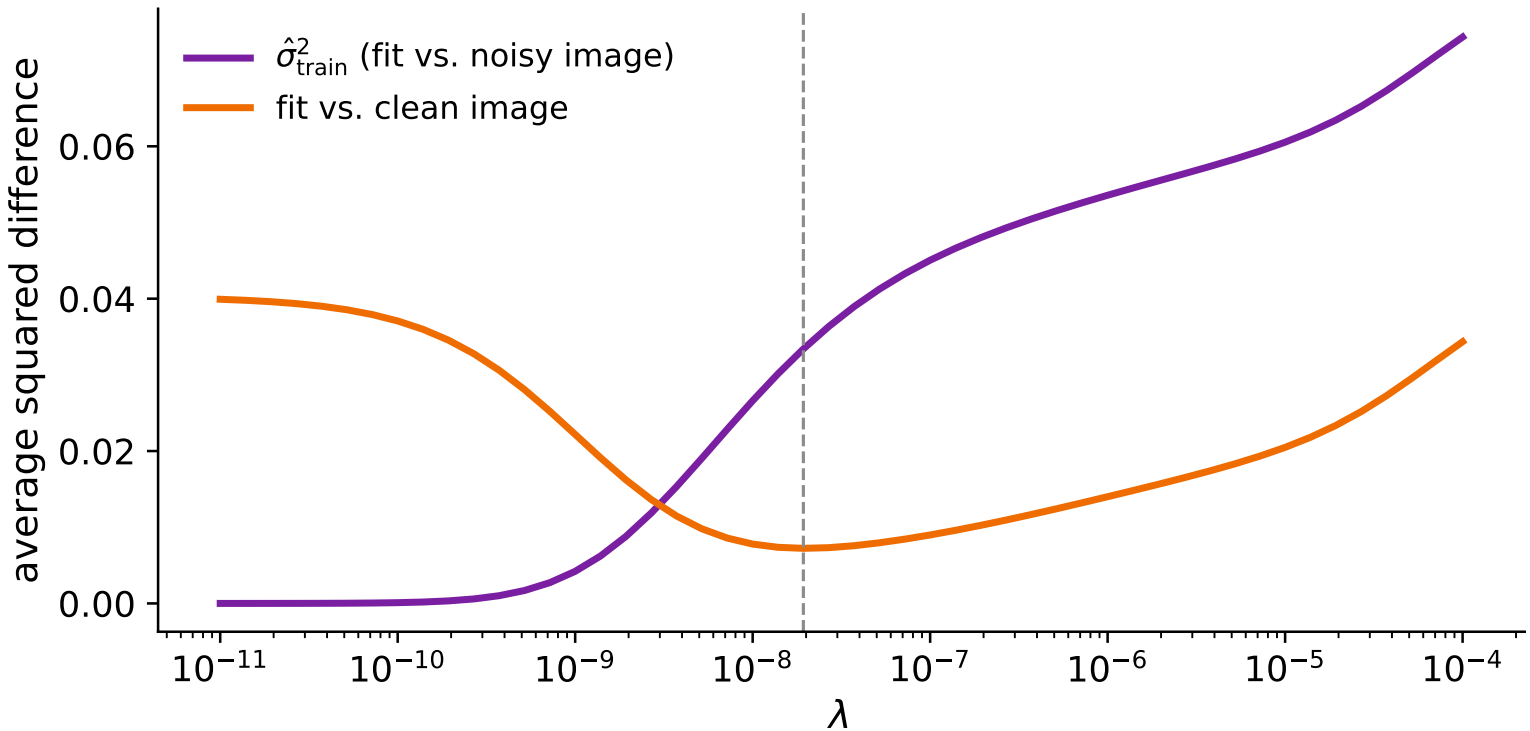


fit, $\lambda = 10^{-5}$



Lena River delta, Siberia, Landsat 7, 27 July 2000. Image: NASA Goddard Space Flight Center.

Choosing λ for the image



Solutions

1) Cancer treatment. Without drug A the options are no drugs (0), C alone ($\beta_C = 2$), B alone ($\beta_B = 3$), and B and C ($\beta_B + \beta_C + J_{BC} = 3 + 2 - 4 = 1$). The best is drug B alone, with an expected reduction of 3 mm. Adding C to B makes it worse because of the interaction $J_{BC} = -4$.
2) $R^2 \approx 1 - \hat{\sigma}^2_{\text{train}}/\hat{\sigma}^2_Y = 1 - 0.5/2 = 0.75$.
3) (a) $\hat{\sigma}^2_{\text{train}}$ decreases (it can never increase, since the larger model can always set the new coefficient to 0) and reaches 0 once the polynomial passes through every training point. $\hat{\sigma}^2_{\text{test}}$ first decreases and then increases, the U shape, and becomes very large as K approaches N_{train}.
   (b) No. The variance term $E_X\left[\text{var}_D(\hat{y}(X, D))\right]$ is an average over X. A large average does not mean the standard error of $\hat{y}(x, D^{\text{train}})$ is large at every x.
   (c) No. $E_D[\hat{\sigma}^2_{\text{test}}] = E_X\left[\text{MSE}_{\hat{y}(X,D)}\right] + \sigma^2_\epsilon \geq \sigma^2_\epsilon$.